Chapter 2

Transport and Deposition of Aerosol Particles

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2.1 INTRODUCTION

This chapter reviews the theoretical models available to describe particle transport in typical semiconductor processing environments. Recent or classic references have been provided wherever possible for additional background. In all of the discussion that follows, particle concentrations are assumed to be low enough so that the influence of the particle on fluid transport can be neglected; particle–particle interactions are also neglected. Under this assumption, the fluid and thermal fields are calculated first (in the absence of particles), and then used as input for subsequent particle transport calculations. The theoretical underpinnings for both the Lagrangian approach (where individual particle trajectories are calculated) and the Eulerian approach (where the particle concentration field is modeled as a continuum) are presented. The strength of the Lagrangian formulation is in predicting particle transport resulting from external forces including particle inertia; but the current implementation cannot describe the chaotic effect of particle Brownian motion (i.e., particle diffusion) on particle transport. On the other hand, the Eulerian formulation can describe particle transport resulting from applied forces and particle diffusion, but the current implementation cannot account for particle inertia.

This chapter begins with a discussion of noncontinuum effects which play a key role in the transport of small particles at low pressure. Next follows a description of the Lagrangian particle transport equation, which basically describes how a particle responds when external forces are applied to it. Brief summaries of some of these forces (which are likely to be important inside a semiconductor process tool) are also included. The concept of particle relaxation time is then introduced, which leads to a discussion of particle drift velocity. The chapter includes a description of the Eulerian particle transport equations for predicting diffusion contributions to particle deposition and concludes with a detailed analysis of particle transport in a parallel plate reactor as an illustration of the utility of these models.

2.2 NONCONTINUUM CONSIDERATIONS

In the following discussion, frequent mention will be made of the continuum and free molecular regimes. These terms are used here to distinguish between the two limiting cases characterizing the nature of the particle/gas interaction.[a] In the continuum limit (large particles or high gas pressures), the gas surrounding the particle appears as a continuous fluid and traditional continuum fluid dynamics apply—such as the Navier–Stokes equations for fluid motion. In the free molecular limit (small particles or low gas pressures), however, the discrete nature of the gas becomes important and individual molecule/particle collisions must be considered. Discrimination between these two regimes is made by comparing the particle diameter to the gas mean free path (which is defined as the average distance a molecule travels between collisions with other gas molecules); a dimensionless parameter known as the Knudsen number is commonly used for these comparisons:

$$Kn = \frac{2\lambda}{d_{\mathrm{p}}} \tag{2.1}$$

where

- λ gas mean free path (cm) $= \mu/(\phi\rho\bar{c})$
- d_{p} particle diameter (cm)
- μ gas viscosity (g cm^{-1} s^{-1})
- ρ gas density (g/cm^3) $= PM/RT$ (for ideal gas)
- $\bar{c}$ mean thermal velocity of the gas molecules (cm/s) $= (8RT/\pi M)^{1/2}$
- R universal gas constant (8.31451×10^7 g cm^2 s^{-2} K^{-1} mol^{-1})
- M gas molecular weight (g/mol)
- P gas pressure (dyne/cm^2 or g cm^{-1} s^{-2}) (1 atm $=$ 760 torr $= 1.01325 \times 10^6$ dyne/cm^2)
- T gas temperature (K)

Also, ϕ is a dimensionless parameter that depends on the kinetic-theory model used to define the gas mean free path: in this work the value $\phi = 0.491$ has been adopted.[1] At atmospheric pressures, the mean free path is typically less than 0.1 μm. Gas mean free path is inversely proportional to pressure at constant temperature; for example, the mean free path in air is 0.674 μm at 76 torr, 6.74 μm at 7.6 torr, and 67.4 μm at 760 mtorr at 296 K. Thus, for low-pressure applications, the Knudsen number for submicron particles can be large.

A large Knudsen number (say >10) corresponds to the free-molecular regime, while a small Knudsen number (say <0.1) corresponds to the continuum regime. Typically, verified theoretical expressions are available in the literature for the forces acting on particles in both the continuum and free-molecular limits. Unfortunately, theoretical force expressions are difficult to formulate in the transition regime that lies between the continuum and free-molecular regimes (particle size of the order of the mean free path, $Kn = 1$). Instead, interpolating or correlating functions are used which go to both the continuum and free-molecular expressions in the limit and match experimental data (if available) in between.

2.3 LAGRANGIAN PARTICLE EQUATION OF MOTION

In the Lagrangian approach to particle transport, particle Brownian motion is neglected and individual particle trajectories (position and velocity as a function of time) are determined by integrating the following system of ordinary differential vector equations:

$$\frac{\mathrm{d}x_\mathrm{p}}{\mathrm{d}t} = V_\mathrm{p} \tag{2.2}$$

$$m_\mathrm{p}\frac{\mathrm{d}V_\mathrm{p}}{\mathrm{d}t} = F_\mathrm{D} + F_\mathrm{G} + F_\mathrm{T} + F_\mathrm{E} + \sum F_\mathrm{i} \tag{2.3}$$

where x_p is the particle position vector, V_p is the particle velocity vector, m_p is the particle mass, and F_D, F_G, F_T, F_E, and F_i are the fluid-drag, gravitational, thermophoretic, electric, and any additional forces (diffusiophoresis, wall drag, etc.) acting on the particle. The forces explicitly listed in Eq. (2.3) are those of the greatest interest in analysing parallel-plate tools (Section 2.7); additional forces can be added linearly as needed. A brief review of these forces follows (for greater detail see Refs. 2–4). Because of the low pressures (typically less than 100 torr) and small particle sizes (diameters typically less than one micron) of interest in semiconductor processing, particles are usually sufficiently smaller than the gas mean free path, λ, so that the free-molecular limit of the various force expressions apply. The free-molecule assumption is well justified for particle Knudsen numbers ($Kn = 2\lambda/d_\mathrm{p}$) larger than 10, and is acceptable (accurate within about 20%) for Knudsen numbers as small as two. For Knudsen numbers less than two, force expressions that extend into the transition or continuum regime should be used.

2.3.1 Fluid–Particle Drag Force

A particle moving at a different velocity than the surrounding gas will experience a gas resistance or fluid-drag force. A great deal of research has been devoted to describing fluid-drag; only a brief review of this body of literature is reported here. For a rigid sphere of diameter d_p moving at constant velocity V_p through a fluid with local velocity U and viscosity μ, the drag force is given by (e.g., [5], p. 105):

$$F_D = -\frac{3\pi\mu d_p}{C(Kn)}\left(V_p - U\right)\cdot C_D)\left(Re_p\right)\frac{Re_p}{24} \tag{2.4}$$

where $C_D(Re_p)\cdot Re_p/24$ is a non-Stokesian correction for fluid inertial effects and $C(Kn)$ is a slip correction factor for noncontinuum effects. A fairly simple correlating equation for the non-Stokesian correction has been suggested by Turton and Levenspiel[6]:

$$C_D\left(Re_p\right)\frac{Re_p}{24} = 1 + 0.173Re_p^{0.657} + \frac{0.01721\cdot Re_p}{1 + 16{,}300\cdot Re_p{}^{-1.09}} \tag{2.5}$$

where the particle Reynolds number is given as

$$Re_p = \frac{\rho d_p\left|V_p - U\right|}{\mu} \tag{2.6}$$

and

$$\left|V_p - U\right| = \left[\left(u_p - u\right)^2 + \left(v_p - v\right)^2 + \left(w_p - w\right)^2\right]^{1/2} \tag{2.7}$$

where u, v, and w are the x, y, and z components of velocity for the fluid (without subscript) and the particle (with subscript p). This correlation applies for particle Reynolds number up to about 200,000. Note that only moderate particle Reynolds numbers (say 10 at most) are expected for most semiconductor process applications, so that the far right terms in Eq. (2.5) are small compared to unity. In fact, for submicron particles in typical processing environments, the slip-corrected Stokes drag law [Eq. (2.4) with $C_D(Re_p)\cdot Re_p/24 = 1$] can generally be used with negligible error.

It has already been explained that for small particles or at low gas pressures (large value of Kn), the continuum approximation eventually breaks down and the molecular nature of the gas must be considered. Cunningham[7] was the first to propose that a correction factor (called the Cunningham or slip correction factor), $C(Kn)$, be included in Eq. (2.4) to account for noncontinuum effects. The functional form first suggested by Knudsen and Weber[8] is in common use:

$$C(Kn) = 1 + Kn\left(\alpha + \beta\cdot\exp\left(-\frac{\gamma}{Kn}\right)\right) \tag{2.8}$$

where α, β, and γ are the parameters that depend on the nature of the gas-particle interaction at the particle surface, and so are affected by both gas composition and particle surface roughness. For example, Ishida[9] used a Millikan apparatus to determine the coefficient α for oil-drops in nine common gases. His results were recently re-evaluated by Rader[10] using modern, more accurate values for the electric charge and gas properties; the corrected values for α are given in Table 2.1.

Based on theoretical considerations[11] and experimental data,[12] Rader[10] recommends that the expression $\alpha+\beta=1.647$ can be used to accurately calculate β for most gas and particle–surface combinations; β values calculated by this formula using Ishida's measured values for α are given in Table 2.1. The choice of the third constant, γ, is more difficult as there are no theoretical techniques for estimating it, and as complete data for empirically determining it are limited. Rader[10] fits the available data for air, carbon dioxide, and helium and found γ values of 0.78, 0.92, and 2.0, respectively. For other gases, Rader[10] suggests $\gamma=0.85$. Note that the correlation for the slip correction factor given by Eq. (2.8) is not very sensitive to the value of γ used. In fact, only small errors typically result if the slip factor is calculated using the fitted constants for oil-droplets in air ($\alpha=1.207$, $\beta=0.440$, and $\gamma=0.78$) for different gases or for particles of different surface roughness. A plot of the $C(Kn)$ against Kn is given in

TABLE 2.1 Gas Properties. Density, Viscosity, Molecular Weight, Mean Free Path, and Slip Correction Constants (for Oil Drops) for Nine Common Gases at $T=296.15$ K and $P=760$ torr[10]

Gases	μ_0 (poise)	M (g mol^{-1})	ρ_0 (g cm^{-3})	λ_0 (mm)	α (–)	β (–)	γ (–)
Air	183.47(−6)	28.966	1.192(−3)	0.0674	1.207	0.440	0.78
Ar	224.80(−6)	39.948	1.645(−3)	0.0703	1.227	0.420	–
He	197.11(−6)	4.003	0.165(−3)	0.1943	1.277	0.370	2.00
H_2	88.61(−6)	2.016	0.826(−3)	0.1240	1.141	0.506	–
CH_4	110.75(−6)	16.043	0.661(−3)	0.0545	1.154	0.493	–
C_2H_6	93.37(−6)	30.069	1.251(−3)	0.0333	1.254	0.393	–
i-C_4H_{10}	75.06(−6)	58.123	2.406(−3)	0.0193	1.186	0.461	–
N_2O	147.88(−6)	44.013	1.818(−3)	0.0493	1.207	0.440	–
CO_2	148.12(−6)	44.010	1.823(−3)	0.0438	1.150	0.497	0.92

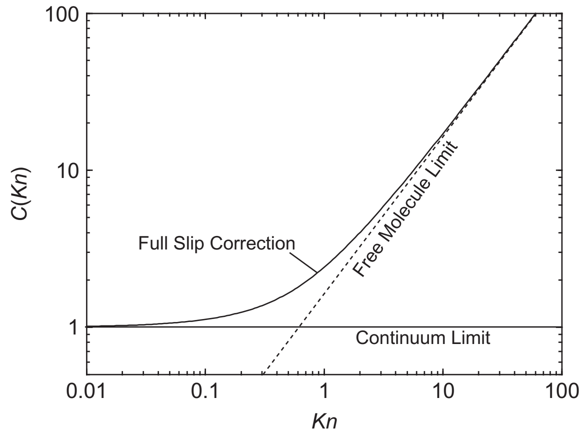

FIGURE 2.1 Slip correction factor. Dependence of the slip correction factor on particle Knudsen number (solid curve). The free molecule limit for the slip correction factor is also shown (dotted curve).

Figure 2.1 using these constants. Note that $C(Kn)$ approaches unity in the continuum limit, and approaches $(\alpha+\beta)Kn$ in the free molecule limit.

2.3.1.1 Fluid-Drag Force: Assumptions and Practical Considerations

All of the above formulae assume solid, homogeneous, spherical particles, while real particles may be far from spherical in shape (flakes, rods, deformed droplets, etc.) and may be porous or inhomogeneous in composition. These variations from ideality can result in particle rotation, modifications to the drag law, etc. Two methods are commonly used to account for nonsphericity: the use of an equivalent diameter or the correction of the drag law with a dynamic shape factor. Hinds[4] and Fuchs[13] provide discussion of both of these approaches. Techniques are becoming available for more accurately modeling the transport of certain classes of nonspherical particles (including flakes, rods, chains of spheres, etc.), but these greatly increase the complexity of the problem.

In adopting Eq. (2.4), resistance terms resulting from fluid inertia (e.g., the virtual mass and Basset history integral terms) have been neglected, which has been shown[13,14] to be a reasonable approximation for aerosol particles (where the particle density is much larger than the fluid density) at particle Reynolds numbers not exceeding a few hundred. The uncertainty in Eq. (2.4) becomes greater at higher Reynolds numbers, but high Reynolds numbers are not expected in semiconductor applications.

Equation (2.4) strictly applies to the uniform (nonaccelerating), straight-line motion of a sphere in a quiescent fluid. In the present application, however, the drag on an accelerating particle in a nonuniform flow field is needed.

Fuchs[13] (Chapter 3) and Clift et al.[14] (Chapter 11) review the issues related to the nonuniform rectilinear motion of aerosol particles. Although it will introduce some inaccuracy, the instantaneous drag acting on an accelerating particle can be estimated with the above constant-velocity drag expression with the fluid and particle velocities taken as their local, instantaneous velocities.

Strictly speaking, the combination of the slip $C(Kn)$ and non-Stokesian $(C_D(Re_p)\cdot Re_p/24)$ corrections in Eq. (2.4) as a product is on flimsy grounds. Henderson,[15] for example, presents a correlation that adds the corrections.[b] Unfortunately, there is little or no data available for $Re \sim 1$ where slip is appreciable, so the proper formulation cannot be decided. For typical semiconductor process environments, however, particle Reynolds numbers are likely to be low so that the non-Stokesian correction is near one and Eq. (2.4) is likely to be quite satisfactory.[c] Thus, in the rest of this work, the slip-corrected Stokes drag law (Eq. (2.4) with $C_D(Re_p)\cdot Re_p/24 = 1$) will be used.

2.3.1.1.1 Continuum Regime Limit

In the continuum limit (large particles and/or near-atmospheric process pressures) and assuming non-Stokesian effects can be neglected, Eq. (2.4) for the gas resistance reduces to Stokes law (e.g., see [4], p. 41):

$$F_{\mathrm{D,\ continuun}} = -3\pi \mu d_p \left(V_p - U\right) \tag{2.9}$$

The continuum regime drag force is seen to be directly proportional to particle size and to the velocity difference between the particle and the gas; it is independent of process pressure (the dependence of fluid viscosity on pressure is very weak) and depends on temperature only through the temperature dependence of viscosity.

2.3.1.1.2 Free Molecule Regime Limit

In the free molecular limit (small particles and/or low process pressures) and assuming non-Stokesian effects can be neglected, Eq. (2.4) for the gas resistance reduces to the following free molecular result (similar to that originally derived by Epstein[16]) in the limit of large Kn:

$$F_{\mathrm{D,\ molecular}} = -\frac{3\pi\phi}{2(\alpha+\beta)}\rho\bar{c}d_p^2\left(V_p - U\right) \tag{2.10}$$

As in the continuum limit, free-molecular drag force is directly proportional to the gas-particle velocity difference. Unlike the continuum result, however, free-molecular drag shows a much stronger (squared) dependence on particle diameter. Another difference is that free-molecular drag is directly proportional to process pressure (through the fluid density term). Temperature dependencies arise implicitly through the fluid density and the mean gas velocity.

2.3.2 Gravitational Force

The gravitational force acting on a spherical particle is given by Hinds ([4], p.42):

$$F_{\mathrm{G}} = \frac{\pi}{6} d_{\mathrm{p}}^3 \left(\rho_{\mathrm{p}} - \rho\right) g \tag{2.11}$$

where ρ_{p} and ρ are the particle and gas density, respectively, and g is the gravitational acceleration vector. The gas density is typically much smaller than the particle density, and can be neglected in Eq. (2.11). The gravitational force is independent of the gas mean free path.

2.3.3 Thermophoretic Force

Because of the thermophoretic force, particles suspended in a gas with a temperature gradient will migrate in the direction opposite to the gradient, i.e. away from hot regions and toward cold regions. This phenomenon results in the preferential deposition of particles on a cold wall, and explains the appearance of a particle free zone near a hot wall. A formulation for the thermophoretic force on a spherical particle was developed by Talbot et al.[17] in a review of thermophoresis:

$$F_{\mathrm{T}} = -3\pi \mu\, d_{\mathrm{p}} \nu K_{\mathrm{T}} . \frac{\nabla T}{T} \tag{2.12}$$

where

$$K_{\mathrm{T}} = \frac{2C_{\mathrm{s}}\left(\dfrac{k_{\mathrm{g}}}{k_{\mathrm{p}}} + C_{\mathrm{t}} Kn\right)}{\left(1 + 3C_{\mathrm{m}} Kn\right)\left(1 + 2 \cdot \dfrac{k_{\mathrm{g}}}{k_{\mathrm{p}}} + 2C_{\mathrm{t}} Kn\right)} \tag{2.13}$$

and where K_{T} is the temperature gradient in the gas, T is the mean gas temperature about the particle, $\nu = \mu/\rho$, k_{g} and k_{p} are the gas and particle thermal conductivities,[d] and C_{t}, C_{s}, C_{m} are the thermal creep coefficient, temperature jump coefficient, and velocity jump coefficient, respectively ($C_{\mathrm{s}} = 1.147$, $C_{\mathrm{t}} = 2.20$, and $C_{\mathrm{m}} = 1.146$ are recommended by Batchelor and Shen[18]). Talbot et al.[17] compared their correlation with other experimenters' data over a wide range of Knudsen numbers, and found it agrees with available experimental data to within 20%.

2.3.3.1 Continuum Regime Limit

In the continuum limit of small Kn the thermophoretic force approaches:

$$F_{\mathrm{T,\,continuum}} = -6\pi \mu^2 d_{\mathrm{p}} \cdot \frac{C_{\mathrm{s}} k_{\mathrm{g}}}{k_{\mathrm{p}} + 2k_{\mathrm{g}}} \cdot \frac{\nabla T}{\rho T} \tag{2.14}$$

which depends on temperature, pressure, gas, and particle thermal conductivities, and is proportional to particle diameter.

2.3.3.2 Free Molecule Regime Limit

In this limit, Eq. (2.12) approaches the limit first derived by Waldmann and Schmitt[19]:

$$F_{\mathrm{T,\,molecular}} = -\frac{\pi}{2}\phi\mu\bar{c}d_{\mathrm{p}}^{2}\cdot\frac{\nabla T}{T} \tag{2.15}$$

Note that in this limit the thermophoretic force is proportional to diameter squared and is independent of the particle thermal conductivity.[e]

2.3.4 Electrostatic Force

A charged particle suspended in a region with an electric field E will experience a force ([4], p. 286):

$$F_{\mathrm{E}} = n_{\mathrm{p}}eE = qE \tag{2.16}$$

where n_{p} is the number of elementary charge units e (4.803×10^{-10} esu in cgs) on the particle giving a total charge q. This equation is deceptively simple, in that the determination of either the charge on the particle or the surrounding electric field can be exceedingly challenging (experimentally or theoretically). In a plasma tool, for example, n_{p} and E will change with time and position and may not be independent.

2.4 INERTIAL EFFECTS

Using the particle fluid-drag force relationship (Eq. (2.4) with $C_{\mathrm{D}}(Re_{\mathrm{p}})\cdot Re_{\mathrm{P}}/24=1$), the particle force balance of Eq. (2.3) can be rewritten as:

$$\begin{aligned}\tau\frac{\mathrm{d}V_{\mathrm{p}}}{\mathrm{d}t} &= U - V_{\mathrm{p}} + \frac{C(Kn)}{3\pi\mu d_{\mathrm{p}}}\left(F_{\mathrm{G}} + F_{\mathrm{T}} + F_{\mathrm{E}} + \Sigma F_{\mathrm{i}}\right)\\ &= U - V_{\mathrm{p}} + V_{\mathrm{G}}^{t} + V_{\mathrm{T}}^{t} + V_{\mathrm{E}}^{t} + \Sigma V_{\mathrm{i}}^{t}\end{aligned} \tag{2.17}$$

where we have introduced a particle response or relaxation time

$$\tau = \frac{\rho_{\mathrm{p}}d_{\mathrm{p}}^{2}C(Kn)}{18\mu} \tag{2.18}$$

and the particle drift velocity, V_{i}^{t} which is discussed below in Section 2.5. Particles characterized by small relaxation times respond rapidly to changes in the flow or in the applied forces, while particles with large relaxation times respond slowly to such changes. For large relaxation times, particle transport is dominated by particle inertia.

2.4.1 Nondimensionalization

It is often convenient to nondimensionalize Eqs. (2.2) and (2.17). For this purpose, a characteristic length and velocity must be specified. For example, for parallel plate geometry (see Figure 2.2), we could chose the inter-plate separation S as the characteristic length, and the magnitude of the mean face velocity U_0 as the characteristic velocity, for which Eqs. (2.2) and (2.3) become:

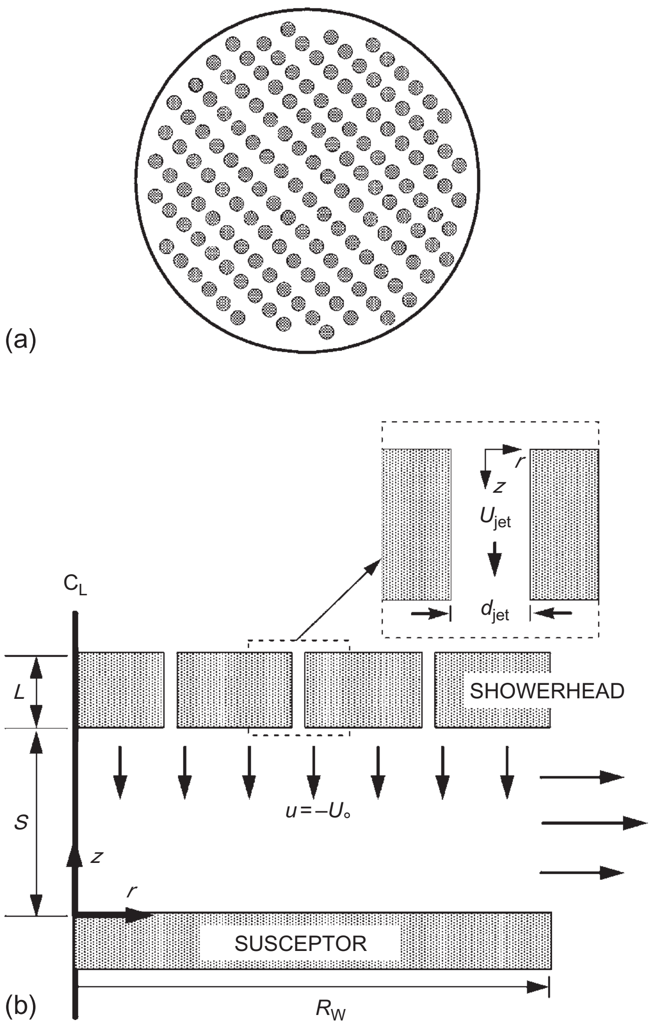

FIGURE 2.2 Parallel-plate reactor geometry. Schematic of the reactor geometry assumed in this work: (a) top view of a showerhead and (b) side view of a parallel-plate reactor.

$$\frac{d\tilde{x}_p}{d\tilde{t}} = \tilde{V}_p \tag{2.19}$$

$$St\frac{d\tilde{V}_p}{d\tilde{t}} = \tilde{U} - \tilde{V}_p + \left(\tilde{V}^t_G + \tilde{V}^t_T + \tilde{V}^t_E + \Sigma\tilde{V}^t_i\right) \tag{2.20}$$

where $\tilde{x}_p = x_p/S$, $\tilde{t} = tU_o/S$, $\tilde{V}_p = V_p/U_0$, $\tilde{U} = U/U_0$, and $\tilde{V}^t_i = V^t_i/U_0$ where V^t_i is the particle drift velocity described below. We have also introduced the particle Stokes number, St, a dimensionless quantity defined as:

$$St = \frac{\tau U_o}{S} = \frac{\rho_p d_p^2 C(Kn) U_o}{18\mu S} \tag{2.21}$$

where τ is the particle relaxation time. The Stokes number is a convenient measure of the importance of particle inertia in a specific reactor; for small St inertial effects can be neglected, while for large St inertial effects must be considered. Physically, the Stokes number can be interpreted as the ratio of the particle stopping distance[f] to the characteristic length of system. The continuum and free molecule limits of St are as follows:

Continuum regime limit:

$$St_{\text{continuum}} = \frac{\rho_p d_p^2 U_o}{18\mu S} \tag{2.22}$$

Free molecule regime limit:

$$St_{\text{molecular}} = \frac{\alpha+\beta}{9\phi}\cdot\frac{\rho_p d_p U_0}{\rho\bar{c}S} = 0.373\cdot\frac{\rho_p d_p U_0}{PS}\cdot\left(\frac{\pi RT}{8M}\right)^{1/2} \tag{2.23}$$

2.5 DRIFT VELOCITY

The manipulation of Eq. (2.3) into Eq. (2.17) resulted in the derivation of the particle drift velocity, V^t_i, which is the particle velocity at which the ith force (neglecting all others) acting on the particle exactly balances the fluid drag force retarding the motion. At this balance point, the particle acceleration vanishes and the particle would move at a steady (or terminal) velocity—the drift velocity. The time required for the particle to reach the drift velocity is short for particles characterized by small particle response times (small Stokes numbers). For small Stokes numbers, particle inertia can be neglected and the acceleration term on the left-hand side of Eq. (2.17) can be dropped, so that particle velocity can be expressed as:

$$V_p = U + V^t_p = U + V^t_G + V^t_T + V^t_E + \Sigma V^t_i \tag{2.24}$$

where the net particle drift velocity is obtained by summing over all external forces, i.e.:

$$V_{\mathrm{p}}^{t}=V_{\mathrm{G}}^{t}+V_{\mathrm{T}}^{t}+V_{\mathrm{E}}^{t}+\Sigma V_{\mathrm{i}}^{t}=\frac{C(Kn)}{3\pi\mu d_{\mathrm{p}}}(F_{\mathrm{G}}+F_{\mathrm{T}}+F_{\mathrm{E}}+\Sigma F_{\mathrm{i}}) \tag{2.25}$$

Equations (2.24) and (2.25) show that, neglecting inertia, the particle will move with the fluid velocity plus the vector sum of the individual drift velocities from all of the applied external forces. In the absence of any external forces (and having neglected particle diffusion), a noninertial particle will exactly follow flow streamlines. For isothermal flow between horizontal, parallel plates, all the V_{i}^{t} are constant and are directed normal to the plates. Expressions are given below for the drift velocities for each of the forces listed in Section 2.3.

2.5.1 Gravitational Drift Velocity

The drift velocity for a spherical particle settling under gravity can be found from Eqs. (2.4) and (2.11) with buoyancy and non-Stokesian effects neglected:

$$V_{\mathrm{G}}^{t}=\frac{\rho_{\mathrm{p}}d_{\mathrm{p}}^{2}gC(Kn)}{18\mu} \tag{2.26}$$

Although the form of the gravitational force is the same in both the continuum and free molecule regime limits, expressions for the two drift velocity limits differ due to the differing drag law contribution:

Continuum regime limit:

$$V_{\mathrm{G,\,continuum}}^{t}=\frac{\rho_{\mathrm{p}}d_{\mathrm{p}}^{2}g}{18\mu} \tag{2.27}$$

Free molecule regime limit:

$$V_{\mathrm{G,molecular}}^{t}=\frac{\alpha+\beta}{9\phi}\cdot\frac{\rho_{\mathrm{p}}d_{\mathrm{p}}g}{\rho\bar{c}}=0.373\cdot\frac{\rho_{\mathrm{p}}d_{\mathrm{p}}g}{P}\cdot\left(\frac{\pi RT}{8M}\right)^{1/2} \tag{2.28}$$

Note that the settling speed of a particle in the free molecule limit varies directly as particle diameter and inversely with pressure, while in the continuum limit the settling speed varies as diameter squared and is independent of pressure.

Example. The gravitational drift velocity for a unit density spherical particle as a function of particle size is shown in Figure 2.3 for six different process pressures in argon at 293 K. For pressures below 100 torr and particle diameters below 1 μm, note that the lines are parallel and straight with a slope of one, as predicted by the free molecule regime limit. Thus, for most of the pressures and particle sizes of interest in semiconductor processing, the free molecule regime limit—Eq. (2.28)—can be used to predict particle settling rates, greatly simplifying calculations.

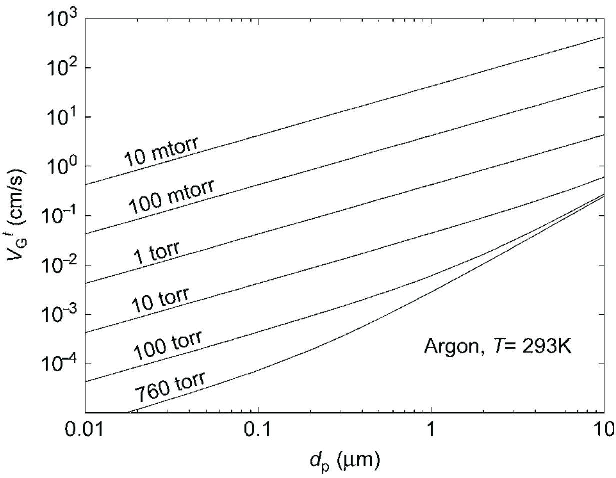

FIGURE 2.3 Gravitational deposition velocity. Dependence of the gravitational deposition velocity on particle diameter for six process pressures (argon at 293 K, particle density $\rho_p = 1$ g/cm^3).

2.5.2 Thermophoretic Drift Velocity

The thermophoretic drift velocity resulting from the balance of thermophoretic and drag forces (neglecting non-Stokesian effects) alone can be found by equating Eqs. (2.4) and (2.12):

$$V_T^t = -K_T C(Kn)\frac{\mu\nabla T}{\rho T} = -K_T C(Kn)\frac{\mu R\nabla T}{PM} \tag{2.29}$$

Interestingly, the drift velocity given by Eq. (2.29) depends on particle diameter only implicitly through the slip correction factor and K_T; specifically, in both the large and small particle limits the thermophoretic velocity becomes independent of particle size.

Continuum regime limit

$$V^t_{T,\,\text{continuum}} = -\frac{2C_S k_g}{k_p + 2k_g}\frac{\mu\nabla T}{\rho T} = -\frac{2C_S k_g}{k_p + 2k_g}\frac{\mu R\nabla T}{PM} \tag{2.30}$$

The continuum limit for thermophoretic drift velocity does not depend on particle size.

Free molecule regime limit

In the free molecular limit, the thermophoretic drift velocity reduces to:

$$V^t_{T,\,\text{molecular}} = -\frac{C_S(\alpha+\beta)}{3C_m}\frac{\mu\nabla T}{\rho T} = -\frac{C_S(\alpha+\beta)}{3C_m}\frac{\mu R\nabla T}{PM} = -0.549\frac{\mu R\nabla T}{PM} \tag{2.31}$$

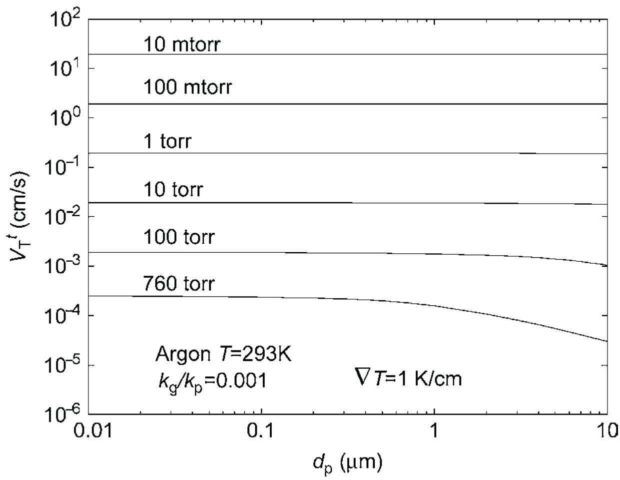

FIGURE 2.4 Thermophoretic deposition velocity. Dependence of the thermophoretic deposition velocity on particle diameter for six process pressures (argon at 293 K, $k_g/k_p = 0.001$, $\nabla T = 1$ K/cm).

which is the same result as given by Waldmann and Schmitt.[19] The free molecule limit is independent of particle size and gas/particle thermal conductivities. The numerical constant 0.549 in the final equality is very general as the sum $\alpha + \beta$ is nearly constant for most gases and particle surfaces.

Example. The thermophoretic drift velocity for a spherical particle as a function of particle size and pressure is shown in Figure 2.4 for an assumed temperature gradient of $\nabla T = 1$ K/cm. For the calculations, the ratio of gas to particle thermal conductivities was taken as $k_g/k_p = 0.001$, representative of a metal particle suspended in argon (the thermophoretic drift velocity is not very sensitive to this ratio). Even for this small temperature gradient, the drift velocity can become large at low pressures. For pressures below 100 torr and particle diameters below 1 μm, note that the drift velocity becomes independent of particle diameter and inversely proportional to process pressure, as predicted for the free molecule regime limit. Thus, for most of the pressures and particle sizes of interest in semiconductor processing, the free molecule regime limit—Eq. (2.31)—can be used to predict particle thermophoretic deposition rates. Because of the simple relationship of drift velocity to pressure and T, calculating the drift velocity for other pressures and temperature gradients is easily done in the free molecule regime limit.

2.5.3 Electric Drift Velocity

The drift velocity for a spherical particle moving under an applied electric field is given by:

$$V_{\mathrm{E}}^{t}=\frac{C(Kn)}{3\pi\mu d_{\mathrm{p}}}\cdot qE \tag{2.32}$$

Although the electrical force is independent of process conditions such as temperature and pressure, these quantities enter the expression for drift velocity through the drag-law contribution.

2.6 EULERIAN FORMULATION

While the Lagrangian (particle tracking) method predicts particle transport by considering single particle motion, the Eulerian formulation predicts particle transport by viewing the particle concentration field as a continuum. In this case, the solution of the particle transport problem becomes very much like that posed by the flow field, i.e., there is one continuous equation for particle mass (concentration) conservation and one continuous equation for particle momentum (velocity) conservation for each particle size. Particle transport by diffusion (Brownian motion) is naturally included in this formulation. A great simplification is obtained if particle inertia is neglected, i.e., it is assumed that the particle instantaneously reaches the drift velocity where drag and imposed forces are in balance.[g] In this case, the particle momentum equation is no longer needed, and only the particle continuity equation for particle concentration n (particles/cm^3) remains ([13], p. 190; 20):

$$\frac{\partial n}{\partial t}+U\cdot\nabla n=\nabla\cdot D\nabla n-\nabla\cdot\left(V_{\mathrm{p}}^{t}n\right)+\Lambda \tag{2.33}$$

where D (cm^2/s) is the Stokes–Einstein particle diffusion coefficient (discussed below) and Λ (#/cm^3/s) is a particle source/sink term to account for particle generation/consumption. The net drift velocity, V_{i}^{t}, appearing in Eq. (2.33) is the same one discussed previously. Although we will not make further use of the fact, it is interesting to note that Eq. (2.33) is applicable to either the Brownian motion of an individual particle or to the diffusion of a particle cloud taken from the continuum point of view ([13], p. 191). For a single particle, n is interpreted as the probability of finding a particle at position (x, y, z) at time t given that the particle was initially located at position (x_0, y_0, z_0) at time t_0. Thus, although semiconductor applications are likely characterized by very low particle concentration levels, the continuum approach can still be applied if we continue to associate the particle concentration with a probability distribution (for example, we may find particle concentrations less than 1 cm^{-3}, which is acceptable from a probabilistic point of view). For boundary conditions, it is assumed that particles which contact the wall stick and are thus instantly removed from the gas, so that the concentration n equals zero at all walls.

Only one-way coupling between the fluid flow and particle concentration fields is used in this work; i.e., the flow field is coupled to particle transport through the velocity field U which appears in Eq. (2.33), while the influence of the particle phase upon the flow is neglected. In practice, the flow field is calculated first (in the absence of a particle phase) and the resulting velocity field is supplied to Eq. (2.33) as a known solution.

2.6.1 Particle Diffusion Coefficient

A more complete discussion of the Stokes–Einstein particle diffusion and its derivation is available in any aerosol text (e.g., [13], Chapter 5). The diffusion coefficient for a spherical particle is:

$$D = \frac{kTC(Kn)}{3\pi\mu d_\mathrm{p}} \tag{2.34}$$

where $C(Kn)$ is defined by Eq. (2.8). The validity of Eq. (2.34) rests on several assumptions: (1) the particles move independently of one another and (2) the movements of a particle in consecutive time intervals are independent.[16] The latter assumption is met only if the condition $t >> \tau$ holds true; in other words, the expression for the diffusion coefficient given in Eq. (2.34) is only valid for observation times much longer than the particle response time.

Continuum regime limit

The continuum regime limit for the diffusion coefficient is

$$D_\mathrm{continuum} = \frac{kT}{3\pi\mu d_\mathrm{p}} \tag{2.35}$$

which is inversely proportional to particle diameter and independent of pressure.

Free molecule regime limit

In the free molecular limit, the diffusion coefficient reduces to

$$D_\mathrm{molecular} = \frac{(\alpha+\beta)}{3\phi}\left(\frac{RT}{2\pi M}\right)^{\frac{1}{2}}\frac{kT}{Pd_\mathrm{p}^2} \tag{2.36}$$

which is inversely proportional to particle diameter squared and pressure.

Example. The particle diffusion coefficient for a spherical particle as a function of particle size and pressure is shown in Figure 2.5 for a temperature of 293 K. For pressures below 100 torr and particle diameters below 1 μm, note that the lines are parallel and straight with a slope of negative two, as predicted for the free molecule regime limit. Thus, for most of the pressures and particle sizes of interest in semiconductor processing, the free molecule regime limit—Eq. (2.36)—can be used to calculate the particle diffusion coefficient.

2.6.2 Nondimensional Formulation

For generality, Eq. (2.33) can be nondimensionalized by choosing a characteristic length (taken here as S, the distance between the showerhead and wafer, Figure 2.2), velocity (taken here as U_0, the mean inlet velocity of the flow at

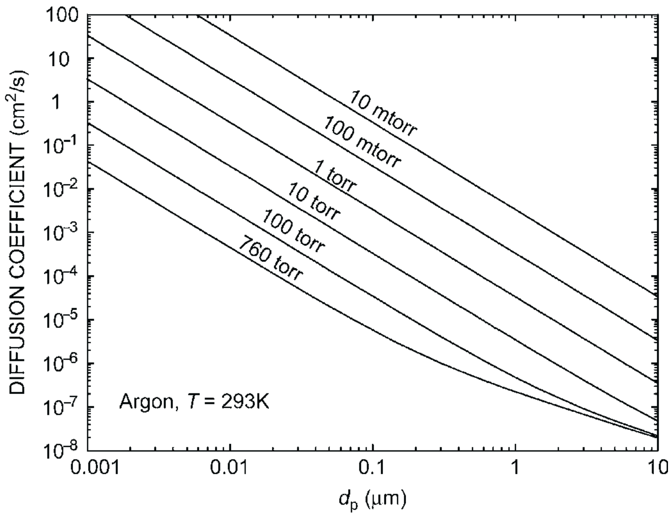

FIGURE 2.5 Diffusion coefficient. Dependence of the particle diffusion coefficient on particle diameter for six process pressures in argon at 293 K.

the showerhead, Figure 2.2), and concentration ($n_0 = \Lambda_h / U_0$, the trap source strength divided by inlet velocity). For steady state, and assuming a constant diffusion coefficient, Eq. (2.33) can be written as[20]

$$\widetilde{U} \cdot \nabla \tilde{n} = \frac{1}{Pe} \nabla^2 \tilde{n} - \nabla \cdot \left(\widetilde{V}_{\mathrm{p}}^{t} \tilde{n} \right) + 1 \tag{2.37}$$

where the Peclet number (the ratio of convective to diffusive transport) is defined as

$$Pe = \frac{SU_0}{D} \tag{2.38}$$

The *continuum regime limit* for the Peclet number is

$$Pe_{\mathrm{continuum}} = \frac{3\pi \mu d_{\mathrm{p}} SU_0}{kT} \tag{2.39}$$

which is proportional to particle diameter. Thus, in the continuum limit, the Peclet number can be used as a dimensionless particle diameter.

In the *free molecular limit:*

$$Pe_{\mathrm{molecular}} = \frac{3\phi}{(\alpha + \beta)} \left(\frac{2\pi M}{RT} \right)^{\frac{1}{2}} \frac{P d_{\mathrm{p}}^2 SU_0}{kT} \tag{2.40}$$

which is proportional to particle diameter squared and to pressure. In the free molecular limit, the square root of the Peclet number can be used as a dimensionless particle diameter.

In standard problems of species mass transfer, the Peclet number would be sufficient to completely characterize the problem for a given geometry and flow field. For particles, however, the presence of a drift velocity term (the second term on the right-hand side) means that the Peclet number no longer uniquely specifies the solution and a dimensionless drift velocity ratio must also be considered:

$$\widetilde{V}_{\mathrm{p}}^{t} = \frac{V_{\mathrm{p}}^{t}}{U_{\mathrm{o}}} \tag{2.41}$$

2.7 PARTICLE TRANSPORT AND DEPOSITION IN A PARALLEL PLATE REACTOR

To illustrate an application of the particle transport models, this section analyzes particle transport in an enclosed, parallel-plate reactor geometry characteristic of a wide range of single-wafer process tools. The axisymmetric geometry we consider consists of uniform flow exiting a showerhead separated by a small gap from a parallel susceptor, as shown in Figure 2.2. The wafer would rest on the susceptor, but for the present analysis the wafer is assumed to be thin enough to be ignored. The showerhead consists of a material (usually a metal or ceramic) through which a large number of holes are drilled (see Figure 2.2a). As one major function of the showerhead is to evenly distribute the flow across its face, the holes are usually made very small in diameter and are very numerous (hundreds to thousands for an 8 inch wafer process tool). Ideally, a showerhead would produce a flow characterized by a mean axial (or face) velocity that does not vary in the radial direction; such flow uniformity is needed to accomplish uniform deposition or etching of the wafer surface. In practice, however, commercial showerheads are typically designed empirically to improve process parameters (such as uniformity); the resulting showerhead designs often create nonuniform flow fields which compensate for other system deficiencies, such as radial temperature or reactive species gradients. Various flow fields can be obtained by manipulation of showerhead hole sizes, numbers, and positions.

One common feature of showerhead design is that the area available to the flow is constricted inside the showerhead; consequently, the velocity of the gas inside the holes of the showerhead is much larger than the face velocity in the gap below. Particles originating upstream of the showerhead and suspended in the flow can be dramatically accelerated while passing through the showerhead, so that at the exit of the showerhead particle velocities much larger than the fluid face velocity are possible. Depending on conditions, particle acceleration by the showerhead can lead to inertia-enhanced particle deposition on the wafer below.[21] Thus, a complete description of particle deposition on a wafer in a parallel-plate reactor must include a description of particle transport through the showerhead as well as an analysis of particle transport in the inter-plate region.

No attempt is made here to analyze particle generation mechanisms; for the present discussion, particles are assumed to originate either: (1) upstream of the showerhead with a known concentration or (2) from a specified position between the plates with a fixed number or at a known generation rate.

The determination of particle transport in a reactor must always begin with a determination of the fluid flow and temperature fields. Particle concentrations are assumed to be low enough to allow a dilute approximation, for which the coupling between the fluid and particle phases is one way. The fluid/thermal transport equations can be solved either analytically or numerically neglecting the particle phase. The resulting velocity and temperature fields are then used as input for the particle transport calculations.[h] In all of the present work isothermal flow is assumed, although small temperature differences are allowed to drive particle thermophoresis. Both analytical and numerical solutions of the flow field are presented.

To provide a single parameter that can be used to compare particle deposition among many cases, a particle collection efficiency is defined as the fraction of particles that deposit on the wafer. Particles are presumed to either enter the reactor through the showerhead (uniformly spread between $r=0$ and R_W), or to originate in a plane parallel to the wafer. The latter case would correspond to particles being released from a plasma trap upon plasma extinction; in this case, the particles are initially assumed to be uniformly spread radially between $r=0$ and R_W at some distance h from the wafer. Analytical expressions for collection efficiency are presented for the limiting case where external forces control deposition (i.e., neglecting particle diffusion and inertia).

Particle transport is predicted using both a Lagrangian approach (where individual particle trajectories are calculated) and an Eulerian approach (where the particles are modeled as a continuum phase). The strength of the Eulerian formulation is in predicting particle transport resulting from the combination of applied external forces (including the fluid drag force) and the chaotic effect of particle Brownian motion (i.e., particle diffusion), although the current implementation cannot account for particle inertia. In particular, the Eulerian formulation cannot accommodate particle acceleration effects within the showerhead, and is therefore restricted to particle transport in the inter-plate region. The Eulerian formulation yields an analytical description of particle deposition for the case where the flow field between the plates can be approximated analytically with a creeping-flow assumption and where the particles are assumed to originate from a planar trap located between the plates. The Lagrangian formulation can account for inertia-enhanced deposition resulting from particles which originate upstream of the showerhead and which are accelerated while passing through it.

The problem is treated in two steps: (1) within a showerhead-hole and (2) between the showerhead and susceptor.

2.7.1 Fluid Transport Equations

In both of the domains considered (flow within the showerhead and between two parallel plates), the geometry will be axisymmetric. With constant fluid properties, and incompressible, laminar, steady flow, the governing equations for axisymmetric flow are the conservation of mass:

$$\frac{\partial u}{\partial z}+\frac{1}{r}\frac{\partial}{\partial r}(rv)=0 \tag{2.42}$$

and conservation of momentum:

$$\begin{aligned}\rho\left(\frac{\partial v}{\partial t}+v\frac{\partial v}{\partial r}+u\frac{\partial v}{\partial z}\right)&=-\frac{\partial P}{\partial r}+\mu\left(\frac{\partial^2 v}{\partial r^2}+\frac{1}{r}\frac{\partial v}{\partial r}+\frac{\partial^2 v}{\partial z^2}-\frac{v}{r^2}\right)\\ \rho\left(\frac{\partial u}{\partial t}+v\frac{\partial u}{\partial r}+u\frac{\partial u}{\partial z}\right)&=-\frac{\partial P}{\partial z}+\mu\left(\frac{\partial^2 u}{\partial r^2}+\frac{1}{r}\frac{\partial u}{\partial r}+\frac{\partial^2 u}{\partial z^2}\right)\end{aligned} \tag{2.43}$$

where u and v are the axial and radial components of the fluid velocity, P is the pressure, ρ is the fluid density, and μ is the fluid viscosity ([22], p. 85).

Boundary conditions are needed to complete the problem specification. In all of the following, no-slip (zero velocity) conditions are taken at all solid walls (i.e., on the showerhead, susceptor, and walls of the showerhead holes), zero radial velocity is assumed along the centerline, and zero traction is assumed at all outflows. For the two flow domains, specific boundary conditions and methods for solving the governing equations are discussed in greater detail below. Note that for convenience, a different coordinate system is used for describing the flow through a showerhead hole than in the region between the parallel plates (see Figure 2.2b).

The generality of the present results are improved if the fluid equations are solved in nondimensional form. Because there are two domains of interest, there are two choices for a characteristic length and velocity. For flow in the showerhead holes, the hole diameter d_{jet} and the magnitude of the mean velocity $\overline{U}_{\text{jet}}$ are used as the characteristic length and velocity, for which the tube Reynolds number is defined as $Re_{\text{jet}}=\rho\overline{U}_{\text{jet}}d_{\text{jet}}/\mu$. For the flow between two plates, the inter-plate separation S and the magnitude of the mean face velocity U_0 are the appropriate choices, and the inter-plate flow is then characterized by a separate Reynolds number: $Re=\rho U_0 S/\mu$.

2.7.1.1 Flow Field in the Showerhead Holes

The idealized geometry and the coordinate system for fluid and particle transport in the showerhead holes is shown in the inset in Figure 2.2b. As seen, z is taken as increasing in the direction of flow (toward the wafer), so that all fluid and particle axial velocities considered in this part of the solution are positive.

The flow in the showerhead holes is assumed laminar with parallel streamlines, thereby neglecting any axial variations in velocity. For laminar flow in a tube with a uniform inlet velocity, however, it is well known that a fully developed parabolic velocity profile develops over an entrance length given approximately by 0.04 $d_{\text{jet}}\, Re_{\text{jet}}$ (5, p. 72). For many showerheads, this entrance length is much less than the hole length (showerhead thickness) and so may be safely neglected; for thin showerheads, however, this may not be the case. In the present analysis, we consider two limiting velocity profiles that meet the above assumptions: (1) plug flow (constant velocity profile) and (2) fully developed laminar flow (parabolic velocity profile). For laminar flow, the velocity profile anywhere along the hole will fall somewhere between these two limiting cases. In case one, the velocity is constant throughout the tube and, for incompressible flow, is equal to the mean velocity in the hole, $\overline{U}_{\text{jet}} = 4Q/\left(N_{\text{jet}}\pi d_{\text{jet}}^2\right)$ where Q is the total gas volumetric flow rate through the showerhead and N_{jet} is the number of individual holes in the showerhead. By mass conservation, it can be shown that the ratio of the mean axial velocity in a hole and the face velocity is the ratio of the showerhead area to the total hole area:

$$\frac{\overline{U}_{\text{jet}}}{U_{\text{o}}} = \frac{A_{\text{showerhead}}}{\Sigma A_{\text{jet}}} = \frac{D_{\text{W}}^2}{N_{\text{jet}} d_{\text{jet}}^2} \tag{2.44}$$

For case two, fully developed laminar flow in a tube is given by

$$U_{\text{jet}}(r) = 2\overline{U}_{\text{jet}}\left(1 - \left(\frac{r}{a_{\text{jet}}}\right)^2\right) \tag{2.45}$$

where r is the radial distance from the tube centerline and a_{jet} is the radius of the hole. The maximum velocity for parabolic flow is twice the mean velocity and occurs on the centerline.

2.7.1.2 Fluid Transport Between Parallel Plates

Both analytical and numerical techniques have been used to calculate the fluid flow between the showerhead and susceptor. For both methods, we assume axisymmetric, incompressible, constant property, laminar, steady flow between two parallel plates (Figure 2.2b). The flow enters through the showerhead ($z = S$) and is assumed to spread immediately, so that the inlet boundary condition is assumed to be a uniform axial velocity, $-U_0$, with zero radial velocity. Both the radial and axial components of velocity vanish at the lower plate ($z = 0$). The analytical approach assumes that the plates are infinite in the radial direction and that the Reynolds number is small; the numerical technique is used for finite plates and is valid for higher Reynolds numbers. The numerical technique is relied on here to define the conditions over which the simpler analytical solution is valid.

The assumption of constant-property flow requires further comment. Although the numerical methods used here can solve for coupled fluid-thermal transport, we have not used this capability in the present analysis. For cases where temperature differences are large or where more accurate solutions are needed for a specific application, the reader is advised to solve the coupled fluid-temperature problem. As mentioned above, a modest temperature gradient is allowed to drive particle thermophoresis (the magnitude of all particle drift velocities will be calculated using fluid properties evaluated at the susceptor temperature).

Under the above assumptions, the flow between the plates is entirely determined by the geometry and the Reynolds number $Re = \rho U_0 S/\mu$. For the range of conditions encountered in semiconductor reactor processes, the associated Reynolds numbers are typically less than 1, and seldom greater than 10. For small Reynolds numbers, viscous effects dominate fluid inertial effects and the "creeping flow" or Stokes flow regime is encountered; it is this regime that allows an analytical solution. The numerical method used for calculating the fluid velocity field was the commercial fluid dynamics analysis code FIDAP (Version 7, Fluid Dynamics International, Evanston, IL, USA). The numerical technique was used to calculate flow fields for Reynolds numbers up to eight.

With fixed values for plate separation ($S = 1$), mean inlet velocity ($U_0 = 1$), and fluid viscosity ($\mu = 1$), the fluid density ρ was varied to obtain flow field solutions for Reynolds numbers between one and eight. The FIDAP option of solving the Stokes flow equations ($Re = 0$) was also used. The results of these calculations are given in Figure 2.6, which shows axial and radial velocity profiles at $r = 1$ for $Re = 0$, 1, 2, 4, and 8 as a function of the dimensionless axial coordinate z/S. Note that all velocities have been normalized by the magnitude of the inlet velocity U_0, and that the radial velocity is also normalized by radius. For $Re = 0$, the radial velocity profile is found to be parabolic and symmetric around $z/S = 0.5$. As Reynolds number increases, the symmetry vanishes and the maximum in the radial velocity moves closer to the plate ($z/S = 0$). Variations in the axial and radial velocity profiles are seen to be quite small for Reynolds numbers less than two. As in the previous work in a similar geometry,[26] it was found that the flow was quasi-1-D: the axial velocity is independent of radius, while the radial velocity is found to scale with radius such that v/r is independent of radius.

An analytical simplification is gained if the flow between the plates can be approximated as a quasi-1-D stagnation point flow. Terrill and Cornish[23] give an asymptotic solution to the problem of axisymmetric, laminar, incompressible, constant property and steady flow between two co-axial infinite parallel disks with constant injection across the disks (a uniform gas inlet velocity across the showerhead). Under these assumptions, a similarity solution reduces the 3-D Navier–Stokes equations to a system of ordinary differential equations; for low Reynolds numbers, these equations can be solved with a power series in Reynolds number.[23]

LIVE GRAPH
Click here to view

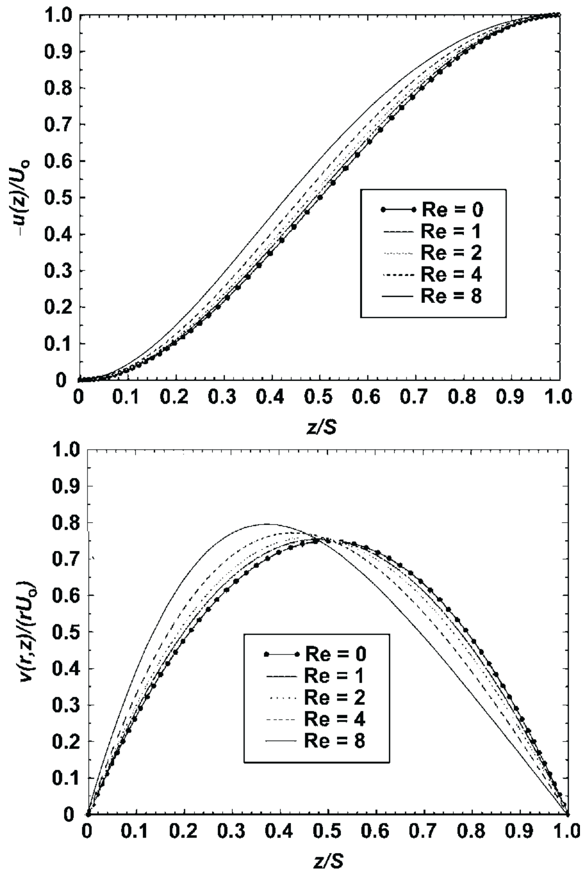

FIGURE 2.6 Flow field results for various Reynolds numbers. Axial (a) and radial (b) velocity profiles for $Re=0$, 1, 2, 4, and 8 calculated on a refined (30 elements) mesh ($r/S=1$ for all curves).

The first two terms of their asymptotic expansion (translated into the present problem definition) are:

$$\begin{aligned} \tilde{u}(\tilde{z}) &= \frac{u(\tilde{z})}{U_o} = 2\tilde{z}^3 - 3\tilde{z}^2 + \frac{Re}{70}\left(2\tilde{z}^7 - 7\tilde{z}^6 + 18\tilde{z}^3 - 13\tilde{z}^2\right) \\ \tilde{v}(\tilde{r},\tilde{z}) &= \frac{v(\tilde{r},\tilde{z})}{U_o} = \tilde{r}\left[3\tilde{z} - 3\tilde{z}^2 - \frac{Re}{70}\left(7\tilde{z}^6 - 21\tilde{z}^5 + 27\tilde{z}^2 - 13\tilde{z}\right)\right] \end{aligned} \tag{2.46}$$

where $\tilde{z}=z/S$ and $\tilde{r}=r/S$. These two equations exactly satisfy all boundary conditions. The quasi-1-D nature of the result is clearly seen as the axial velocity is independent of radius, while the radial velocity scales linearly with radius.

In the limit of vanishingly small Reynolds number, Eq. (2.46) reduces to a symmetric, parabolic profile for radial velocity, in excellent agreement with the

Stokes flow solution ($Re = 0$) obtained by FIDAP (this limit has also been previously reported by Houtman et al.[24]).

Equation (2.46) does a very good job of approximating the axial velocity profile—agreeing with FIDAP solutions to better than 1% for Reynolds numbers less than 4, and to better than about 4% for Reynolds numbers up to 8. The success of Eq. (2.46) in predicting radial velocity is not nearly so good. Although the error is better than 1% for $Re < 1$, the maximum observed error quickly grows, reaching 15% for $Re = 4$ and 70% at $Re = 8$. As can be seen, the largest errors are found near the showerhead ($z/S = 1$), where the magnitude of the radial velocity is quite small. Thus, although the relative error is quite large, the absolute error is small. In any case, our treatment of the region near the showerhead is only approximate because we have neglected the effect of the discrete jets issuing from the showerhead. Strictly from a fluid velocity point of view, Eq. (2.46) provides a very good approximation of the flow for Reynolds numbers less than two, and a reasonable approximation up to a Reynolds number of four.

2.7.1.3 Summary: Fluid Flow Analysis for the Parallel Plate Geometry

This section has defined the parallel-plate geometry which will be used to approximate the flow inside a showerhead-type etch or CVD reactor. The acceleration of the gas flow as it passes through the showerhead will later be found to play a key role in enhancing particle deposition by particle inertia; for this reason, solutions for flow within the showerhead holes have been presented. The two limiting cases which were considered, plug and fully developed parabolic flow, should bracket the range of flows likely to be encountered in semiconductor applications. Laminar, incompressible, constant-property flow between two infinite, parallel plates was used to approximate the inter-plate flow in real reactors which are certainly more complicated. Reynolds number and edge effects were discussed, and an analytical solution was found that should provide a fairly accurate description of the flow for Reynolds numbers less than about four when the plate separation is much smaller than the overall system radial dimension. Variable temperature effects have not been considered.

2.7.2 Particle Collection Efficiency

Particle collection efficiency is defined as the fraction of particles present in the inter-plate region that deposit on the wafer. The collection efficiency is introduced to provide a single parameter that can be used to compare particle transport and deposition results among many cases. Note that the use of collection efficiency side-steps the important issue of the particle source term. Thus, while the present transport analysis addresses the question of the *fraction* of gas-borne particles that deposit on the wafer, a prediction of the *number* of particles that deposit on the wafer additionally requires a clear understanding of the controlling particle generation mechanisms. In practical terms, the present analysis helps identify strategies for reducing the probability that particles are transported to and deposit on a wafer; a complete strategy for reduction of total particle-on-wafer counts also requires that particle source terms be understood and controlled.

Three particle-source scenarios are considered: (1) a continuous source of particles entering the inter-plate region through the showerhead with known concentration (such as for contaminated process gases), (2) a discrete number of particles that are originally trapped between the plates (such as by a plasma) but are subsequently released (such as at plasma extinction), and (3) a continuous source of particles which are created between the plates at a known generation rate (such as by particle nucleation). General collection efficiency expressions for these cases are defined below. In addition, analytical expressions are provided for the limiting case where external forces control particle deposition—i.e., both particle inertia and Brownian motion are neglected. In the absence of particle inertia and Brownian motion, Robinson[25] has shown that particle concentration is constant along particle trajectories if: (1) the flow is incompressible and (2) the external forces acting on the particle are all divergence free. For the infinite parallel-plate geometry with constant-property flow, the flow is clearly incompressible and the second condition is met for the gravitational and Coulombic electric particle forces (which are each constant between the plates).

2.7.3 Particles Entering Through the Showerhead

For this case, collection efficiency is defined as the fraction of particles entering the inter-plate region through the showerhead (between $r=0$ and R_W) that deposit on the wafer. These particles are presumed to originate upstream of the showerhead and are assumed to be evenly distributed across the showerhead. Particle acceleration through the showerhead will be considered in Section 2.8.1. The present Lagrangian formulation accounts for the coupling between particle inertia and external forces in determining particle transport in the inter-plate region. The calculation of a collection efficiency with a Lagrangian technique requires the determination of the critical radius, R_{crit}, which is the starting radial position (at the showerhead) of a particle that follows a trajectory that leads it to deposit at the edge of the wafer, R_W (see Figure 2.7).

All particles starting closer to the centerline will deposit on the wafer, while those starting farther out will exit the reactor. For a uniform concentration across the showerhead, the collection efficiency, η, can be written as

$$\eta = \left(\frac{R_{crit}}{R_W}\right)^2 = \left(\frac{R_i}{R_f}\right)^2 \tag{2.47}$$

In general, the critical trajectory must be found by a trial-and-error method. The second equality of Eq. (2.47) is a simplification that only applies under our

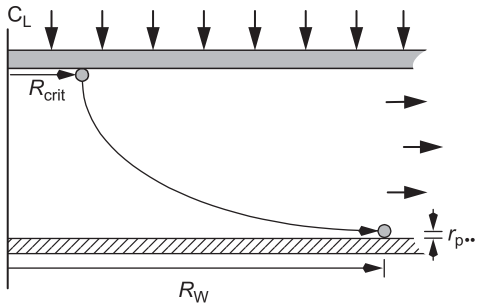

FIGURE 2.7 Critical trajectory. Diagram of a critical trajectory for a particle which starts at the showerhead at radial position R_{crit} and deposits at the wafer edge R_W.

quasi-1-D approximation. In this case, all the factors that influence particle deposition (e.g., axial fluid velocity profile, the particle initial velocity, and particle axial drift velocity) are independent of radial position; thus, the question of whether a particle will hit the wafer must not depend on its initial radial position, R_i (although the *radial* position at which the particle hits the wafer, R_f, will depend on R_i). It can be shown for our quasi-1-D case that the ratio R_i/R_f is independent of initial radial position. Thus, efficiency in the Lagrangian framework is calculated by starting the particle at a particular radial position ($R_i = 1$) and calculating its trajectory to determine the radial position of contact with the wafer[i]; the efficiency is then $(R_i/R_f)^2 = (1/R_f)^2$ as given in Eq. (2.47). The total number of particles depositing on the wafer is the product of efficiency times the total flux of particles entering through the showerhead.

2.7.3.1 External Force Limit

For the case where external forces control particle deposition (neglecting inertia, interception, and diffusion), Rader et al.[26,27] used a Lagrangian analysis to obtain the following expression for deposition efficiency in isothermal, quasi-1-D parallel-plate flows:

$$\eta = \left(\frac{R_{crit}}{R_W}\right)^2 = \left(\frac{R_i}{R_f}\right)^2 = \frac{\left|V_p^t\right|}{\left|V_p^t\right| + U_0} \quad V_p^t \leq 0 \tag{2.48}$$

where V_p^t is the z-component of the net particle drift velocity (the resultant of all external forces in the axial direction).[j] For net drift velocities greater than zero (net external force pushing particles away from the wafer), no particle deposition on the wafer is predicted (although it will be shown later that particle inertia or diffusion can cause deposition even in this case). Equation (2.48) provides a lower bound for particle deposition, as inertial and diffusional effects can only increase deposition from what is predicted. Interestingly, as the particle net drift velocity is typically much smaller than the fluid entrance velocity, Eq. (2.48) predicts that (in the absence of inertia and diffusion) the particles which land on the wafer originate from near the reactor centerline.

Equation (2.48) is independent of the flow field (consider that the flow Reynolds number does not appear) that can be more easily understood by applying Robinson's[25] result (as discussed above). Neglecting diffusion and inertia, the concentration over the lower plate must equal the inlet concentration, n_0. The rate at which particles deposit on the lower plate becomes $|V_p^t| n_o \pi R_W^2$, while the rate at which particles enter through the showerhead is $\left(|V_p^t| + U_o\right) n_o \pi R_W^2$. Taking the ratio of these expressions gives the same efficiency as Eq. (2.48) (see also 28). Similar results, including Eq. (2.48), were found by Ramarao and Tien[29] for plane-stagnation flow.

Interception effects were neglected in Eq. (2.48), which implies that a particle is collected only when its center of mass reaches the wafer surface. A better assumption is that particle collection occurs when the particle comes within one particle radius ($r_p = d_p/2$) of the wafer surface.[k] The derivation of Eq. (2.48) can be easily modified to include interception, with the following result:

$$\eta = \left(\frac{R_{crit}}{R_W}\right)^2 = \left(\frac{R_i}{R_f}\right)^2 = \frac{\left|V_p^t| + |u\left(z = r_p\right)\right|}{|V_p^t| + U_o} \quad V_p^t \le \left|u\left(z = r_p\right)\right| \tag{2.49}$$

Note that particle collection is now expected in the absence of external forces (or even for weak repulsive forces); the physical interpretation is that collection occurs when the flow brings the particle within one particle radius of the wall. The inclusion of interception also has the effect that Eq. (2.49) [unlike Eq. (2.48)] depends on the flow field through the term $u(z = r_p)$. Because the gas velocity one particle radius away from the wafer is typically vanishingly small, interception effects are generally neglected in the following discussion, and Eq. (2.48) is used.

2.7.4 Particle Traps/in Situ Nucleation

Another source of wafer contamination is from particles that start somewhere between the plates, and are subsequently transported to the wafer. One example is particles generated in situ by nucleation. A second example is particles that are originally trapped between the plates during a plasma process, which are subsequently released at plasma extinction. While the plasma is on, contaminant particles generally accumulate in specific regions of the radio frequency (RF) discharge. Roth et al.[30] first used laser light scattering to observe that particles accumulate near the bulk plasma-sheath boundary in these discharges. Sommerer et al.[31] and Barnes et al.[32] first proposed that particle transport in the discharge is dominated by two forces: electrostatic and viscous ion drag. The electrostatic force accelerates negatively charged particles toward the

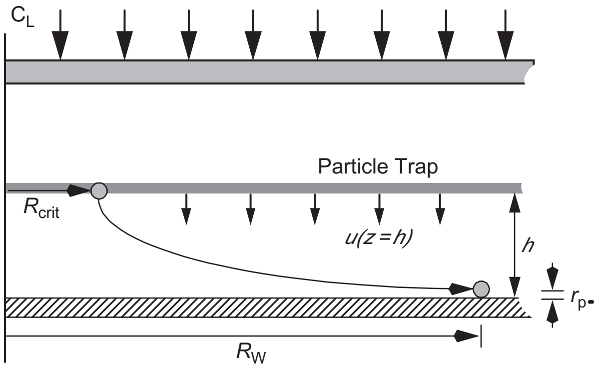

FIGURE 2.8 Trap schematic. Diagram of particles in a planar trap located at a distance h from the lower plate; a critical trajectory is also shown for a particle which starts at radial position R_{crit} and deposits at the wafer edge R_W.

center of electropositive plasmas, while viscous ion drag accelerates particles in the direction of net ion flux (generally toward plasma boundaries). Particle "traps" occur in regions where the sum of forces acting on the particle vanishes. In many cases these traps are approximately planar and parallel to the plates[33]; a schematic of a planar trap is shown in Figure 2.8, where the particles are uniformly distributed at a distance h above the lower plate. Only planar traps are considered in this work, although a variety of other trap structures (rings, domes, etc.) are well known in the literature. For any trap structure more complicated than an infinite plane the problem becomes inherently 2-D, which is beyond the scope of the present analysis.

At the end of the process step, the discharge is extinguished and the plasma-induced forces responsible for particle trapping are assumed to dissipate rapidly (compared to particle transport times) in the afterglow. In this work, we assume that the charged particles are rapidly neutralized after the plasma extinction and can therefore be treated as neutral particles as experimentally observed by Jellum et al.,[34] Shiratani et al.,[35] and Yeon et al.[1,36] Under the assumption of rapid neutralization, the particles are released from the traps and can deposit on the wafer as a result of external forces, inertia, or Brownian motion (diffusion). To analyze the extent of deposition both the Lagrangian and Eulerian formulations have been used. Although the physical interpretation of efficiency (fraction of particles starting in the trap that end up on the wafer) is the same for both approaches, the methods of calculating the efficiency are quite different.

2.7.4.1 Efficiency for the Lagrangian Formulation

In the Lagrangian formulation, Brownian motion is neglected and calculation of particle trajectories is determined from the coupling between particle inertia and external forces. Consequently, the determination of a collection efficiency reduces to the determination of a critical trajectory just as defined in Eq. (2.47) of the previous section, except that the particle starting position is now at axial position h and the particle initial velocity is assumed to be zero. As before, it can be shown for our quasi-1-D case that the ratio R_i/R_f is independent of initial radial position within the trap. Thus, efficiency in the Lagrangian framework is calculated by starting the particle at a particular radial position ($R_i = 1$) in the trap ($z = h$) and calculating its trajectory to determine the radial position of contact with the wafer; the efficiency is then calculated by Eq. (2.47).

2.7.4.2 Efficiency for the Eulerian Formulation

For small particles and/or at low pressure, the effects of Brownian motion on particle transport must be considered. Brownian motion results from random variations in the force exerted on the particle by background-gas molecular bombardment, and gives rise to particle diffusion along concentration gradients. Also, Brownian motion implies that particle trajectories are no longer deterministic; that is, identical particles started at the same initial location with the same initial conditions will not follow the same path through the reactor. In this case, an Eulerian formulation of particle transport is used, in which the particles are treated as a continuum or cloud and the particle concentration field is calculated (inertia is neglected). Particle deposition is determined in terms of a particle flux at the wafer's surface, J_0 (#/cm^2/s), which is calculated from the surface concentration gradient (where particle interception is neglected):

$$J_o = D\frac{dn}{dz}\bigg|_{z=0} \tag{2.50}$$

where n is the particle concentration and D is the particle diffusion coefficient. Note that a general expression for particle flux would include both a diffusional term, given by Eq. (2.50), and a drift-velocity term, given by $n\widetilde{V}_p^t$. In Eq. (2.50) only the diffusional term is shown because, under our assumption that particle concentration vanishes at surfaces, the drift-velocity contribution must also vanish at the susceptor. Thus, even when external forces are controlling deposition, a thin boundary layer must exist near the susceptor wherein the concentration drops from the free-stream value to zero at the susceptor's surface. The particle collection efficiency is then calculated as the ratio of particle flux to the wafer divided by the particle source term (number of particles being released from the trap).

Equation (2.50) can be extended to account for particle interception by evaluating the concentration derivative at r_p (instead of zero):

$$J_o = D\frac{dn}{dz}\bigg|_{z=r_p} \tag{2.51}$$

2.7.4.3 External Force Limit

As in the previous section, an analytical result can be derived for deposition efficiency in the limiting case where external forces control particle deposition (particle inertia, interception, and diffusion are all neglected):

$$\eta = \left(\frac{R_{\mathrm{crit}}}{R_{\mathrm{W}}}\right)^2 = \left(\frac{R_{\mathrm{i}}}{R_{\mathrm{f}}}\right)^2 = \frac{\left|V_{\mathrm{p}}^t\right|}{\left|V_{\mathrm{p}}^t\right| + \left|u(z=h)\right|} \quad V_{\mathrm{p}}^t \le 0 \tag{2.52}$$

which is the same as Eq. (2.48) except that the gas axial velocity at the trap location replaces the mean gas velocity in the denominator. For net drift velocities greater than zero (net external force pushing particles away from the wafer), no particle deposition on the wafer is predicted. Equation (2.52) provides a lower bound for particle deposition, as inertial and diffusional effects can only increase deposition from what is predicted. As expected, the collection efficiency tends toward unity as the particle trap moves closer to the lower plate (h_0) because the axial gas velocity must approach zero at the plate surface. For particles which ultimately deposit on the wafer, Equation (2.52) also can be used to determine the radial position on the wafer at which particles are collected, R_{f}, based on starting position $r = R_{\mathrm{i}}$ and $z = h$. As discussed in the previous section, particles which deposit on the wafer are those which start nearest to the reactor centerline. It should be noted that both the Eulerian and Lagrangian collection efficiencies defined above must tend to Eq. (2.52) in the limit when particle diffusion, inertia, and interception effects are all negligible.

Equation (2.52) can be extended to include particle interception as in the previous section:

$$\eta = \left(\frac{R_{\mathrm{crit}}}{R_{\mathrm{W}}}\right)^2 = \left(\frac{R_{\mathrm{i}}}{R_{\mathrm{f}}}\right)^2 = \frac{\left|V_{\mathrm{p}}^t\right| + \left|u\left(z=r_{\mathrm{p}}\right)\right|}{\left|V_{\mathrm{p}}^t\right| + \left|u(z=h)\right|} V_{\mathrm{p}}^t \le 0 \tag{2.53}$$

As before, particle collection is now predicted in the absence of (or for weak) external forces, and is seen to depend on the flow field through flow velocity terms in both the numerator and denominator. Both the Eulerian and Lagrangian collection efficiencies defined above must tend to Eq. (2.53) in the limit where particle diffusion and inertial effects are negligible.

2.7.5 Diffusion-Enhanced Deposition from Traps or in Situ Nucleation

One difference between these two particle–source scenarios is that the source term resulting from nucleation is continuous, while the source term for a plasma-trap release is a transient event characterized by the number of particles in the trap at the time of release. In either case, the particles of interest are likely to be quite small and chamber pressures may be low, so that the effect of particle Brownian motion must be considered. Although these very small particles are not currently considered to reduce yield, the trend toward smaller feature sizes on integrated circuits is continually reducing the size of a killer defect. Thus, the industry will inevitably be faced with the need to understand the role of diffusion in particle transport and deposition.

The analysis of this section closely follows the previous work of Peters et al.[20] who investigated the diffusive deposition of particles onto disks in an infinite stagnation point flow (such as for a wafer exposed to the downward flow in a clean room). In their work, Peters and coworkers assume axisymmetric viscous stagnation point flow, while in this work an analytical asymptotic result for flow between two axisymmetric infinite parallel plates is used. Peters et al. also used different particle concentration boundary conditions than in this work: particle concentration was assumed to be zero at the disk and to approach a constant infinitely far away from it. Here, the two plates are considered perfectly absorbing (vanishing particle concentration), and a planar particle source is assumed to be located somewhere between them.

2.7.5.1 Problem Definition

We assume the geometry shown in Figure 2.9: axisymmetric flow between two infinite, parallel plates (a showerhead and a susceptor) separated by a distance S. The effect of jetting out of the showerhead holes is neglected, so that the flow is assumed to be uniformly distributed across the bottom of the showerhead with velocity, U_0. The flow is assumed to be isothermal (constant gas properties), steady, laminar, incompressible, and viscous such that the quasi-1-D analytical result [Eq. (2.46)] can be used.

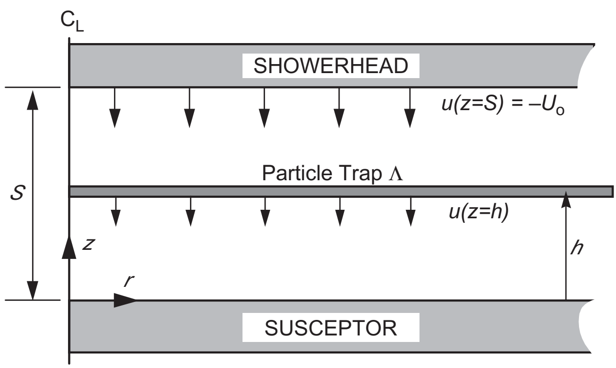

FIGURE 2.9 Geometry. Diagram of parallel, infinite-plate geometry with particles in a planar trap located at a distance h from the lower plate.

As discussed, Eq. (2.46) is reasonably accurate for flow Reynolds numbers less than about four based on comparison to more accurate numerical finite-element simulations.

The particles are assumed to enter the domain at a steady volumetric rate Λ (#/cm^3/s) from a planar source located a distance h from the susceptor. Although this description adequately applies to a continuous source such as particle nucleation, it is not immediately obvious that a steady-state analysis is applicable to the transient case where a finite number of particles are simultaneously released from a trap at time $t=0$. Analysis of the governing equations reveals that the particle collection efficiencies from the steady-state and transient problems are in fact the same under the following conditions: (1) steady flow field, (2) infinite parallel-plate (1-D) domain, (3) radially uniform distribution of initial particle positions for the transient problem, and (4) radially uniform particle source for the steady-state problem. To confirm this contention, particle transport calculations using the present steady-state Eulerian approach have been compared with the Brownian dynamics simulations (BDS) of Choi et al.[38] Choi and coworkers solved the Langevin equation directly using a massively parallel numerical Lagrangian particle tracking model which included a fluctuating Brownian force; transport calculations were presented for particles that were initially distributed in planar traps in a parallel-plate geometry similar to that assumed here. The BDS method is inherently transient in nature, in that a large number of particles were initially distributed uniformly throughout the trap, and their trajectories followed in time until the particles either deposited on a plate or left the calculation domain. For comparison with the present approach, Brownian dynamics simulations were performed with the analytical velocity field given by Eq. (2.46). As expected, BDS results for particle collection efficiency were in excellent agreement with the steady-state Eulerian formulation presented here. Thus, the analytical result for particle collection efficiency given below applies equally well for a steady-state particle planar source as for the case of a cloud of particles released from a planar trap.

2.7.5.2 Solution of the Eulerian Particle Transport Equation

Neglecting particle inertia, the Eulerian expression for particle concentration, n (#/cm^3), is Eq. (2.33) which is reproduced here:

$$\frac{\partial n}{\partial t}+U\cdot\nabla n=\nabla\cdot D\nabla n-\nabla\cdot\left(V_{\mathrm{p}}^{t}n\right)+\Lambda \tag{2.54}$$

where D (cm^2/s) is the Stokes–Einstein particle diffusion coefficient, Λ (#/cm^3/s) is the particle source term, and is the net drift velocity vector. Consistent with our flow assumptions, the concentration field is assumed to be steady and one-dimensional (depending only axial position). Also, for isothermal flow, the diffusion coefficient and particle drift velocity are constant. With these assumptions and simplifications, Eq. (2.33) may be rewritten as

$$u\frac{\mathrm{d}n}{\mathrm{d}z}=D\frac{\mathrm{d}^2n}{\mathrm{d}z^2}-V_\mathrm{p}^t\frac{\mathrm{d}n}{\mathrm{d}z}+\Lambda \tag{2.55}$$

In Eq. (2.55), V_p^t is the z-component of the net drift velocity vector and u is the axial velocity field given by Eq. (2.46). For boundary conditions, the assumption of perfectly absorbing walls is made which implies that the particle concentration is zero at both the upper and lower plates, i.e. $n(z=0)=n(z=S)=0$.[m] Note that by assuming an absorbing surface at the showerhead we are neglecting the showerhead holes, but this is reasonable as the holes typically account for only a few per cent of the total showerhead surface.

For this analysis, the particle source is assumed to be infinitely thin so that

$$\Lambda=\Lambda_h\delta(z-h) \tag{2.56}$$

where Λ_h (#/cm^2/s) is a constant area source term and δ (cm^{-1}) is the Dirac delta function. Although the following derivation also could be followed for a finite-thickness source term, the resulting analytical expression for particle collection efficiency would be much more complicated than that given below.

To nondimensionalize Eq. (2.55), the appropriate characteristic length and velocity scales are S and U_0, respectively. A characteristic particle concentration, n_0, can be defined based on the particle source strength and gas inlet velocity:

$$n_0=\frac{\Lambda_h}{U_0} \tag{2.57}$$

Using these definitions, Eq. (2.55) becomes:

$$\tilde{u}\frac{\mathrm{d}\tilde{n}}{\mathrm{d}\tilde{z}}=\frac{1}{Pe}\frac{\mathrm{d}^2\tilde{n}}{\mathrm{d}\tilde{z}^2}-\widetilde{V}_\mathrm{p}^t\frac{\mathrm{d}\tilde{n}}{\mathrm{d}\tilde{z}}+\delta\left(\tilde{z}-\tilde{h}\right) \tag{2.58}$$

where $\tilde{n}=n/n_0$, $\tilde{u}=u/U_\mathrm{o}$, $\tilde{z}=z/S$, $\widetilde{V}_\mathrm{p}^t=V_\mathrm{p}^t/U_0$, $Pe=SU_0/D$, and $\tilde{h}=h/S$. As discussed above, the solution for the dimensionless concentration is completely determined by the dimensionless groups Pe, $\tilde{h}$, $\widetilde{V}_\mathrm{p}^t$, and Re (which enters implicitly as $\tilde{u}$ depends on Reynolds number).

After defining a dimensionless concentration gradient

$$\tilde{G}=\frac{\mathrm{d}\tilde{n}}{\mathrm{d}\tilde{z}} \tag{2.59}$$

Equation (2.58) can be rewritten as

$$\frac{\mathrm{d}\tilde{G}}{\mathrm{d}\tilde{z}}-Pe\left(\tilde{u}+\widetilde{V}_\mathrm{p}^t\right)\tilde{G}=-Pe\,\delta\left(\tilde{z}-\tilde{h}\right) \tag{2.60}$$

The solution to Eq. (2.60) is

$$\tilde{G} = \tilde{G}_0 \exp(A) - \exp(A) \int_0^{\tilde{z}} Pe\, \delta(\tilde{z} - \tilde{h}) \exp(-A) \mathrm{d}\tilde{z} \tag{2.61}$$

where

$$A(\tilde{z}) = Pe\left[\frac{1}{2}\tilde{z}^4 - \tilde{z}^3 + \frac{Re}{70}\left(\frac{1}{4}\tilde{z}^8 - \tilde{z}^7 + \frac{9}{2}\tilde{z}^4 - \frac{13}{3}\tilde{z}^3\right) + \widetilde{V}_{\mathrm{p}}^t \tilde{z}\right] \tag{2.62}$$

and $\tilde{G}_{\mathrm{o}}$ is the dimensionless concentration gradient at the lower plate, $\tilde{z} = 0$. Note that $\tilde{G}_{\mathrm{o}}$ is frequently referred to as the Sherwood number, Sh (e.g., [20]). To determine $\tilde{G}_{\mathrm{o}}$ apply the boundary conditions $(\tilde{z} = 0) = \tilde{n}\ (\tilde{z} = 1) = 0$ after integrating Eq. (2.59):

$$\int_0^1 \frac{\mathrm{d}\tilde{n}}{\mathrm{d}\tilde{z}} \mathrm{d}\tilde{z} = \int_0^1 \mathrm{d}\tilde{n} = 0 = \int_0^1 \tilde{G} \mathrm{d}\tilde{z} \tag{2.63}$$

Solving Eq. (2.63) for $\tilde{G}_{\mathrm{o}}$ gives

$$\tilde{G}_0 = \frac{Pe \cdot \displaystyle\int_{\tilde{h}}^1 \exp\left\{Pe\left[\frac{1}{2}\left(t^4 - \tilde{h}^4\right) - \left(t^3 - \tilde{h}^3\right) + \frac{Re}{70}\left(\frac{1}{4}\left(t^8 - \tilde{h}^8\right) - \left(t^7 - \tilde{h}^7\right) + \frac{9}{2}\left(t^4 - \tilde{h}^4\right) - \frac{13}{3}\left(t^3 - \tilde{h}^3\right)\right) + \widetilde{V}_{\mathrm{p}}^t\left(t - \tilde{h}\right)\right]\right\} \mathrm{d}t}{\displaystyle\int_0^1 \exp\left\{Pe\left[\frac{1}{2}t^4 - t^3 + \frac{Re}{70}\left(\frac{1}{4}t^8 - t^7 + \frac{9}{2}t^4 - \frac{13}{3}t^3\right) + \widetilde{V}_{\mathrm{p}}^t t\right]\right\} \mathrm{d}t} \tag{2.64}$$

2.7.5.3 Particle Collection Efficiency

The particle collection efficiency is found as the ratio of particle flux to the lower plate divided by the total number of particles entering the reactor:

$$\eta = \frac{A_{\mathrm{w}} D \left.\dfrac{\mathrm{d}n}{\mathrm{d}z}\right|_{z=0}}{A_{\mathrm{w}} \displaystyle\int_0^S \Lambda_h \delta(z - h) \mathrm{d}z} = \frac{\dfrac{D\Lambda_h}{SU_0} \left.\dfrac{\mathrm{d}\tilde{n}}{\mathrm{d}\tilde{z}}\right|_{\tilde{z}=0}}{\Lambda_h} = \frac{\tilde{G}_0}{Pe} \tag{2.65}$$

where A_{w} is the area in the r–θ plane. Thus, an analytical result for the particle collection efficiency is given by Eqs. (2.64) and (2.65)—although the solution requires numerical quadrature. The dependence of the collection efficiency on the four dimensionless groups, Re, $\tilde{h}$, $\widetilde{V}_{\mathrm{p}}^t$, and Pe, is clearly shown in Eqs. (2.64) and (2.65). For comparison to previous work in the literature, the particle collection efficiency defined by Eq. (2.65) can also be expressed as the ratio of the Sherwood to Peclet numbers, $\eta = Sh/Pe$. The use of collection efficiency as the dimensionless number characterizing the deposition process is preferred to the Sherwood number for this application for two reasons: (1) determination of the collection efficiency is also straightforward for Lagrangian formulations to be applied to showerhead-enhanced inertial deposition in which critical trajectories can be calculated and (2) efficiency is a commonly accepted concept within the semiconductor industry (e.g., yield). In practical terms, an efficiency of unity indicates that all particles in the chamber are depositing on the wafer, while an efficiency of zero indicates that no particles are depositing on the wafer.

2.7.5.4 Particle Flux

The particle deposition rate on the wafer is found as the product of collection efficiency times the number of particles released from the trap (or generated by nucleation):

$$J_0 = D\frac{\mathrm{d}n}{\mathrm{d}z} = \frac{D\Lambda_h}{SU_0}\frac{\mathrm{d}\tilde{n}}{\mathrm{d}\tilde{z}}\Big|_{\tilde{z}=0} = \Lambda_h\frac{\tilde{G}_0}{Pe} = \Lambda_h\eta \tag{2.66}$$

When the nature of the source term Λ_h is not known, the best strategy for reducing the number of defects on a wafer is to choose conditions that inhibit particle transport to the wafer, i.e., minimize the collection efficiency. One potential weakness of this strategy is if process conditions selected to reduce the collection efficiency result in a corresponding increase in the particle generation rate; this possibility is certainly of concern for particle nucleation.

2.7.6 Nondimensional Results

This section presents calculations of particle collection efficiency using numerical quadrature of Eqs. (2.64) and (2.65) based on a fourth-order Runge–Kutta technique with automatic error control. Both local and global error control parameters can be set, and convergence tests showed that the resulting integrals were unchanged in the fifth place when these parameters were set to 10^{-10}. Efficiency is found to be a function of four dimensionless parameters: Re, $\tilde{h}$, $\tilde{V}_\mathrm{p}^t$, and Pe (a fifth—the interception parameter d_p/S—is neglected in this work). Calculations of particle collection efficiency versus Peclet number are shown in Figure 2.10 for creeping flow ($Re=0$) for three attractive forces (characterized by $\tilde{V}_\mathrm{p}^t = -0.1$, -0.5, and -1.0), for no external force ($\tilde{V}_\mathrm{p}^t = 0$), and for three repulsive forces ($\tilde{V}_\mathrm{p}^t = 0.1$,0.5, and 1.0). Plots are shown for three different trap heights, where the particles are trapped: (a) near the wafer ($h/S=0.1$), (b) midway between the wafer and showerhead ($h/S=0.5$), and (c) near the showerhead ($h/S=0.9$).

2.7.6.1 Efficiency at Intermediate Peclet Numbers

Although the small- and large-Pe asymptotic limits for the collection efficiency are well described by analytical expressions, the shape of the collection

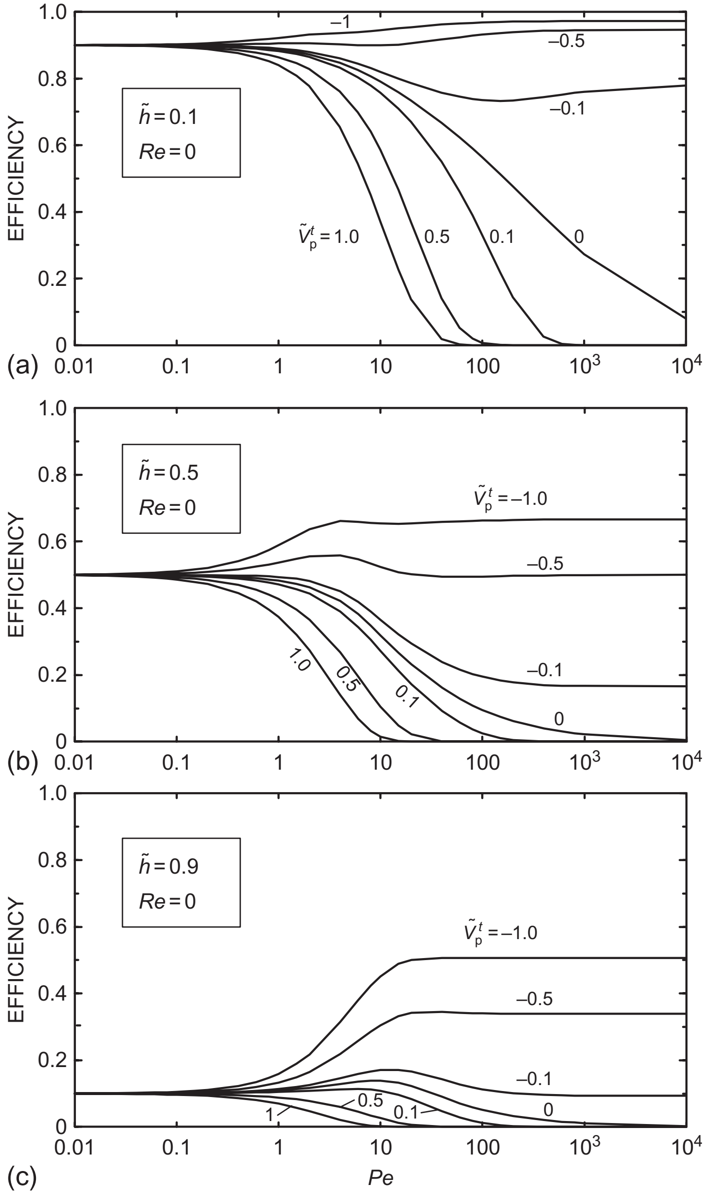

FIGURE 2.10 Efficiency vs. *Pe* for various deposition velocities. Figures show calculated efficiencies for particles starting in traps at: (a) $h/S = 0.1$, (b) $h/S = 0.5$, and (c) $h/S = 0.9$.

LIVE GRAPH
Click here to view

efficiency curves for intermediate Peclet numbers can be quite complex and requires the full numerical integration of Eqs. (2.64) and (2.65). For example, while the efficiency-curve transition between the small- and large-Pe asymptotic limits is generally monotonic (e.g., Figure 2.10a for $\widetilde{V}_{\mathrm{p}}^{t} > 0$ or Figure 2.10c for $\widetilde{V}_{\mathrm{p}}^{t} < -0.5$), in some cases there may be a local minimum (e.g. Figure 2.10a for $\widetilde{V}_{\mathrm{p}}^{t} = -0.1$) or maximum (e.g., Figure 2.10c for $\widetilde{V}_{\mathrm{p}}^{t} = -0.1$). The exact shape of the efficiency curve depends on the magnitudes of the three parameters, $\tilde{h}$, $\widetilde{V}_{\mathrm{p}}^{t}$, and Pe, and although the interaction among them can be complex, a few simple observations can be made. First, moving the particle trap away from the wafer (i.e., increasing $\tilde{h}$) always tends to lower the collection efficiency. Although this effect is most notable for low or intermediate Pe values where diffusional effects are strong, it is also true for large Pe values where deposition is controlled by external forces. The latter claim is supported by noting that gas velocity at the trap location, $\tilde{u}(\tilde{h})$, increases with increasing distance from the wafer so that collection efficiency decreases. Trap manipulation can be accomplished in practice under some conditions. For example, in plasma processing, the trap location is determined by process parameters such as pressure, rf (radio frequency) power, and flow rate; while these parameters may be fixed during etching by process requirements, they could be adjusted to manipulate the particle trap location just prior to plasma extinction. Similarly, trap position when particle nucleation is present could be controlled by pressure, wall temperature, flow rates, or chemistry selections.

Second, reducing the dimensionless attractive external force or increasing the dimensionless repulsive force always lowers the collection efficiency. This trend is clearly evident in Figure 2.10 for intermediate and large Peclet numbers, although the benefit becomes less significant at low Pe where diffusion dominates deposition. It is interesting to note that for attractive forces such as gravity (wafer facing up) or thermophoresis (wafer cooler than the surrounding gas), the dimensionless drift velocity increases as pressure is decreased (for constant U_0). In this case, the tendency toward processing at lower pressures ultimately must lead to an increase in the fraction of particles which end up on the wafer. If a process recipe is selected that maintains the wafer warmer than its surroundings, however, the drift velocity resulting from this repulsive external force will increase with decreasing operating pressure (for constant U_0) and thereby reduce the fraction of particles depositing on the wafer.

To explore the effect of Reynolds number on collection efficiency, calculations were made for Reynolds numbers of 0 and 8 using the analytical approximation for the flow field given in Eq. (2.46)[n]; the results are shown in Figure 2.11. As shown, even this relatively large variation in Re (spanning the Re range of the majority of low pressure commercial tools) produces only modest variations in the collection efficiency. Reynolds number effects are most apparent in the large-Pe limit for attractive forces. This effect can be quantified by noting that: (1) the large-Pe efficiency limit of Eq. (2.52) depends on gas velocity at the trap location, $\tilde{u}(\tilde{h})$, and (2) the value of $\tilde{u}(\tilde{h})$ changes

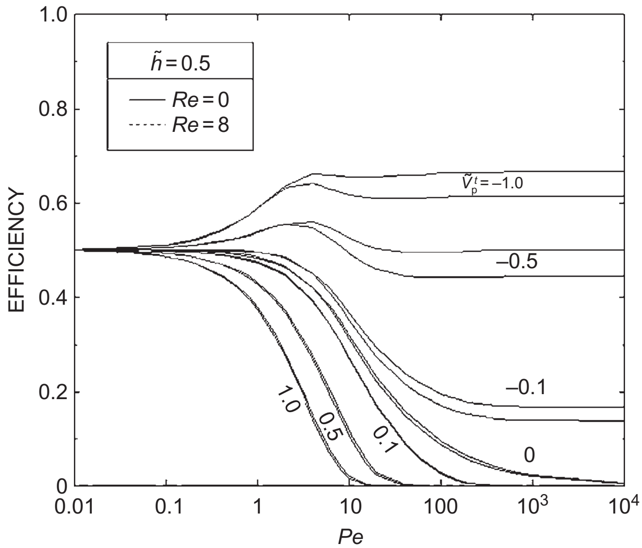

FIGURE 2.11 Efficiency vs. *Pe* for various deposition velocities. Figures show calculated efficiencies for $Re=0$ and 2 ($h/S=0.5$).

from -0.5 at $Re=0$ to a value of -0.625 at $Re=8$. This 25% increase in $\tilde{u}(\tilde{h})$ associated with *Re* being increased from 0 to 8 leads to an approximately 25% decrease in collection efficiency for a weak attractive force, i.e. $|\tilde{V}_p^t| << |\tilde{u}(\tilde{h})|$. As $\tilde{V}_p^t$ is increased to a magnitude comparable to $\tilde{u}(\tilde{h})$, this effect diminishes; for example, for $\tilde{V}_p^t=-1$ the efficiency for $Re=8$ is only 8% less than for $Re=0$.

For small Peclet numbers, the effect of Reynolds number vanishes for all values of the external force. In this case, particle transport by diffusion dominates convective transport, so that the collection efficiency is decoupled from the details of the flow field. Figure 2.11 also shows that Reynolds effects are negligible in the presence of repulsive external forces; although this conclusion is true in an absolute sense, Reynolds number effects do become important in a relative sense for larger Peclet numbers where the collection efficiency becomes small. For example, for $\tilde{V}_p^t=1.0$ and a $Pe=15$, the collection efficiency for $Re=8$ ($\eta=3.18\times10^{-3}$) is approximately 50% larger than that for $Re=0$ ($\eta=2.11\times10^{-3}$). While such differences may not be detectable at low particle concentrations, the difference in the number of particles depositing on the wafer may become quite large when particle concentrations are high—such as is typical of systems in which particle nucleation is occurring.

2.7.7 Dimensional Results

This section presents several example calculations of collection efficiency in dimensional terms when gravity and diffusion act simultaneously. The solution scheme described in the previous section is again used here, except that the dimensional inputs are first converted into the required nondimensional groups before the numerical integrations are performed. The particle diameter replaces the Peclet number as the independent variable in all of the following. All of the examples assume a 200 mm diameter with a showerhead-to-wafer gap of 2.54 cm. For these calculations, a baseline process is taken as argon flowing at a mass flow rate of 1000 sccm (standard cubic centimeters per minute) at a pressure of 1 torr and temperature of 300 K. Constant gas properties are assumed (isothermal flow) along with a particle density of 1 g/cm^3. The Reynolds number for these conditions is 0.984, which indicates viscous dominated flow and is well inside the range for which the analytical flow field expression can be used. In the following examples, trap height, pressure, flow rate, and pressure are individually varied about the baseline value. Note that these parameters may not be independent; for example, trap height may depend on both pressure and gas flow rate. Finally, the section concludes with a demonstration of the reduction in deposition that can be obtained by introducing a force that opposes deposition, such as heating the wafer relative to the showerhead to take advantage of thermophoretic protection.

2.7.7.1 Trap Height Effects

Plots of calculated particle collection efficiency as a function of particle size are shown in Figure 2.12 for dimensionless trap heights of 0.1, 0.3, 0.5, 0.7, and 0.9.

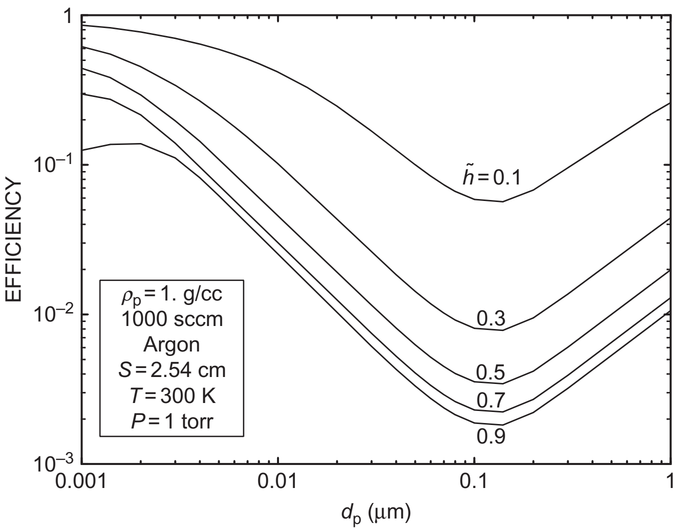

FIGURE 2.12 Efficiency vs. particle diameter and trap position (isothermal case). Collection efficiencies for particle transport over a 200 mm wafer including gravitational settling and diffusion for dimensionless trap locations of 0.1, 0.3, 0.5, 0.7 and 0.9 ($S = 2.54$ cm, $Q = 1000$ sccm argon, $P = 1$ torr, $T = 300$ K, $Re = 0.98$, and particle density $= 1$ g/cm^3).

LIVE GRAPH
Click here to view

All of the curves exhibit a minimum near 0.1 μm, with increasing efficiency for both smaller and larger sizes. This shape has commonly been reported in previous deposition studies: for example, see Figure 3 of[39] which shows the net stagnation-point drift velocity based on additivity of convective-diffusion, electrostatic, and gravitational velocities. In Figure 2.12, the increase in efficiency below 0.1 μm is associated with increasing diffusional deposition, while the increase in efficiency above 0.1 μm is associated with increasing gravitational deposition. Note that in the present geometry, the diffusional branch does not increase without bound, but instead asymptotically approaches the highly diffusive limit $\eta \rightarrow 1-\tilde{h}$. As seen in Figure 2.12, however, this limit is not quite achieved even for particles as small as 0.001 μm. As expected based on the previous discussion in nondimensional terms, the trapping height plays a key role in net particle deposition. Clearly, it is always advantageous to manipulate the particle trap to a location as far from the wafer as possible. It is clear from Figure 2.12 that as the size of IC-killing particles shrinks below 0.1 μm that particle collection efficiencies will climb as a result of increased particle diffusion.

2.7.7.2 *Pressure Effects*

Plots of calculated particle collection efficiency as a function of particle size are shown in Figure 2.13 for reactor pressures of 1, 10 100, and 760 torr. For these calculations, the mass flow rate is held constant at the baseline value of 1000 sccm, and the trap height is assumed to be 1.27 cm which is exactly halfway between the wafer and showerhead. As before, all of the curves show the characteristic "U" shape resulting from the combination of deposition from

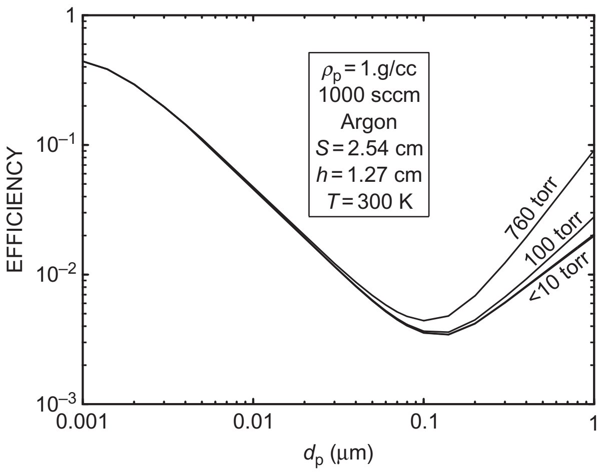

FIGURE 2.13 Efficiency vs. particle diameter and pressure (isothermal case). Collection efficiencies for particle transport over a 200 mm wafer including gravitational settling and diffusion for reactor pressures of 1, 10, 100, and 760 torr (S = 2.54 cm, Q = 1000 sccm argon, h = 1.27 cm, T = 300 K, Re = 0.98, and particle density = 1 g/cm^3).

LIVE GRAPH
Click here to view

convective-diffusion and gravitational settling. It is evident, however, that the diffusional branch of the efficiency curves is independent of pressure for this example. This result is explained by considering that the Peclet number (which along with geometry completely specifies the convective-diffusive problem) depends on the ratio of inlet gas velocity to the particle diffusion coefficient, and that both of these quantities are inversely proportional to pressure at a constant mass flow rate. The pressure dependence of the diffusion coefficient is evident in the free molecule limit given in Eq. (2.36), which applies at low pressures and for small particle size. The velocity used in scaling the problem, U_0, is the lineal gas velocity at the reactor pressure, and for a presumed constant mass flow rate this also must scale inversely proportional to pressure. Thus, the collection efficiency resulting from particle diffusion is nearly independent of pressure for a fixed gas mass flow rate.

The branch of the efficiency curve resulting from gravitational settling is also seen to be independent of pressure below 10 torr. This result is explained by considering the limiting expression for external-force dominated deposition, which depends only on $\tilde{u}(\tilde{h})$ and $\tilde{V}_{\mathrm{p}}^{t}$. The term $\tilde{u}(\tilde{h})$ is independent of pressure as it depends only on trap position and Reynolds number (see Eq. (2.46)), and as at constant mass flow rate the Reynolds number is independent of pressure. In the free molecule limit given by Eq. (2.28), the gravitational drift velocity, $\tilde{V}_{\mathrm{p}}$ varies inversely with pressure so that its ratio to the inlet velocity, $\tilde{V}_{\mathrm{p}}^{t}$, must also be independent of pressure. Thus, all of the terms describing the collection efficiency are independent of pressure in the particle free molecule limit. As pressure increases above 10 torr and for particles larger than 1 μm, however, the free molecule limit for the settling velocity no longer applies and the full expression of Eq. (2.26) must be used. As the particle mean free path decreases, the inverse pressure dependence of the dimensional settling velocity diminishes, until the continuum regime limit Eq. (2.27) is reached which is independent of pressure. As V_{p}^{t} becomes independent of pressure, $\tilde{V}_{\mathrm{p}}^{t}$ becomes proportional to pressure, resulting in the marked increase in collection efficiency with pressure as seen in Figure 2.13 for larger particles and at higher pressures.

2.7.7.3 Mass Flow Rate Effects

Plots of calculated particle collection efficiency as a function of particle size are shown in Figure 2.14 for argon mass flow rates of 10, 100, and 1000 sccm. For these calculations, the reactor pressure is held constant at the baseline value of 1 torr, and the trap height is assumed to be 1.27 cm (half-way between the wafer and showerhead). As before, all of the curves show the characteristic “U” shape resulting from the combination of deposition from convective-diffusion and gravitational settling. These results show that the collection efficiency is a strong function of gas mass flow rate except for very small particle sizes, for which diffusion dominates and all curves must tend toward the same limit. Note that the point at which the diffusion-dominated limit is achieved varies with

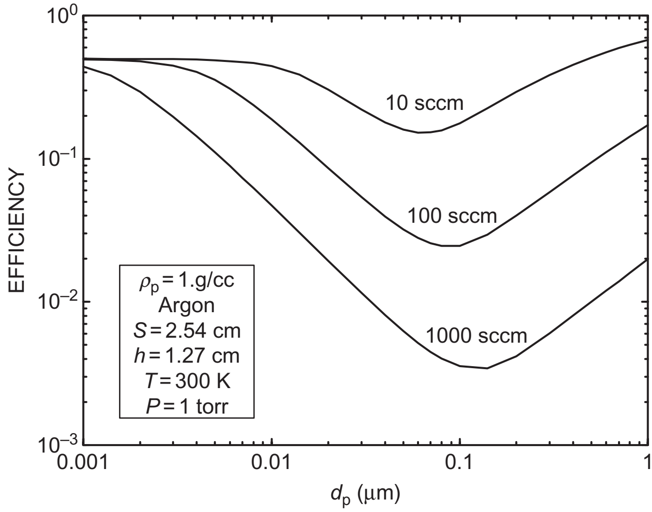

FIGURE 2.14 Efficiency vs. particle diameter and flow rate (isothermal case). Collection efficiencies for particle transport over a 200 mm wafer including gravitational settling and diffusion for gas mass flow rates of 10, 100, and 1000 sccm argon ($S=2.54$ cm, $Q=1000$ sccm argon, $P=1$ torr, $T=300$ K, $Re=0.98$, and particle density $=1$ g/cm^3).

mass flow rate: for the lowest flow rate of 10 sccm the limit ($\eta=0.5$) is reached for particles less than about 0.01 μm, while at the highest flow rate of 1000 sccm the collection efficiency is still below the limit for 0.001 μm. Thus, a significant reduction in particle collection efficiency can be achieved by increasing the mass flow rate at constant pressure. The effect of mass flow rate on the actual *number* of particles depositing on the wafer is less obvious. For example, consider the case in which the mass flow rate is increased from 100 to 1000 sccm. Although the collection efficiency drops by approximately a factor of 10, the flow increases by a factor of 10, so that if the number of particles entering the domain scales with flow rate, then the total number of particles depositing on the wafer should be about the same for the two flow rates. If, however, the number of particles present is independent of the flow rate, then increased flow rates should reduce the number of particles on the wafer.

For calculations where pressure is fixed, it should be noted that the Reynolds number increases proportionately with mass flow. However, even at the highest flow rate considered, 1000 sccm, the Reynolds number is less than one and the analytical flow approximation is excellent.

2.7.7.4 Effect of Thermophoresis

This section explores the role of thermophoresis in determining particle collection efficiency. For these calculations, the showerhead temperature has

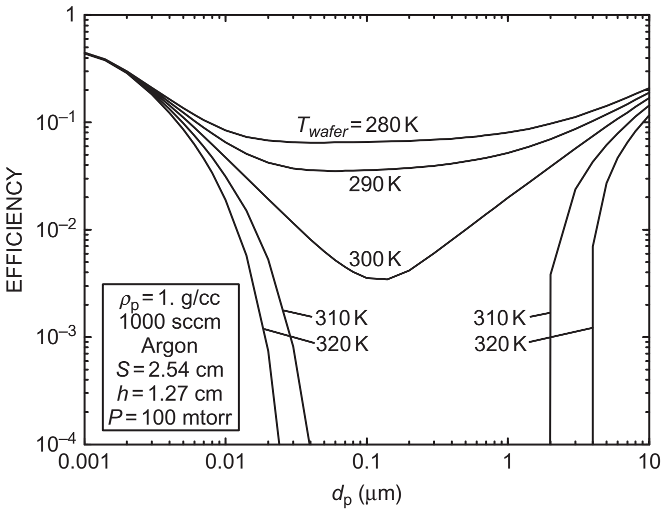

FIGURE 2.15 Efficiency vs. particle diameter and wafer temperature. Collection efficiencies for particle transport, including gravity, thermophoresis and diffusion for wafer temperatures of 280, 290, 300, 310, and 320 K (200 mm wafer, 1000 sccm argon, $S = 2.54$ cm, $h = 1.27$, $P = 100$ mtorr, $T_{showerhead} = 300$ K, and particle density = 1 g/cm^3).

been held constant at 300 K, and the wafer temperature varied to produce a temperature gradient that drives thermophoretic deposition. To accommodate our assumption of constant properties, only small temperature differences are considered. Plots of calculated particle collection efficiency as a function of particle size are shown in Figure 2.15 for wafer temperatures of 280, 290, 300, 310, and 320 K. The baseline conditions described above are used for all of these calculations: reactor pressure of 100 mtorr, argon mass flow rate of 1000 sccm, wafer-to-showerhead gap of 2.54 cm, and the trap height is assumed to be 1.27 cm. A particle of density 1 g/cm^3 was assumed, and the ratio of the gas to particle thermal conductivity was taken as 0.02 to approximate a fused silica particle. As before, all of the curves show the characteristic "U" shape resulting from the combination of deposition from convective-diffusion and external forces, where the net external forces contain contributions from both gravitational and thermophoretic forces. The minimum collection efficiency for the case of thermophoretic protection (wafer hotter than showerhead) becomes vanishingly small and so is not shown; note that even for thermophoretic protection the collection efficiency never reaches absolute zero as there always is some contribution from diffusion/convection.

A key result of these calculations is that even modest temperature differences can lead to dramatic changes in collection efficiency. For example, deposition is nearly eliminated in the 0.1–1.0 μm range when the wafer is kept only 10–20°C warmer than the showerhead. On the other hand, the collection efficiency increases by an order of magnitude when the wafer temperature is decreased 10°C as compared to the isothermal case. In addition, when the wafer temperature is kept below the showerhead temperature, the depth of the collection efficiency minimum at about 0.1 μm becomes shallower; this results from the fact that the thermophoretic drift velocity is nearly independent of particle diameter as indicated by Eqs. (2.30) and (2.31). These calculations clearly demonstrate the importance of keeping the wafer warmer than its surroundings at all times.

2.7.8 Summary: Diffusion-Enhanced Deposition

This section explored particle deposition resulting from external forces and Brownian motion in a parallel-plate geometry characteristic of a wide range of semiconductor process tools. The need to properly account for diffusion-enhanced particle deposition becomes increasingly important as the semiconductor industry moves toward smaller feature sizes and becomes concerned with smaller-sized particles. Particle transport was modeled using the Eulerian approach of Section 2.6, so that the continuum convective-diffusion equation was solved for particle flux. One strength of the Eulerian formulation is in predicting particle transport resulting from the combination of applied external forces (including the fluid-drag force) and the chaotic effect of particle Brownian motion (i.e., particle diffusion), although the current implementation neglected particle inertia. Furthermore, particles were assumed to originate in a planar trap located between the plates, and only transport in the inter-plate region was considered (showerhead acceleration was neglected). Flow between infinite parallel plates was assumed as described by the quasi-1-D creeping flow approximation, where the showerhead was treated as a porous plate.

The key result of the analysis of this section was the derivation of expressions for the particle collection efficiency—which is the fraction of trapped particles which end up on the wafer. An analytical, integral expression was derived that gives the particle collection efficiency as a function of four dimensionless parameters: Re, $\tilde{h}$, $\tilde{V}_p^t$, and Pe (a fifth—the interception parameter d_p/S—was neglected). The first parameter, the Reynolds number, completely specifies the flow field under the present assumptions. The second parameter, the dimensionless trap height, $\tilde{h} = h/S$, specifies the position of the particle source term. The influence of external forces enters through the third parameter, the dimensionless particle drift velocity, $\widetilde{V}_p^t = V_p^t/U_0$ which is defined as the z-component of the net drift velocity. The fourth parameter is the particle Peclet number, $Pe = SU_0/D$, which is a measure of the relative importance of particle Brownian motion. In the free molecular limit, the Peclet number is proportional to diameter squared, so that $Pe^{1/2}$ can be thought of as a dimensionless particle size.

Numerical quadrature of Eqs. (2.64) and (2.65) using a fourth-order Runge–Kutta technique was used to calculate particle collection efficiency in terms of the controlling dimensionless parameters. Initial calculations showed the numerical results to be in good agreement with the various analytical limits, providing confidence in the current implementation. In general, the highly diffusive limit was approached for Peclet numbers less than about 0.1, while the nondiffusing limit was essentially reached for Peclet numbers larger than $\sim 10^2$ or $\sim 10^3$ depending on the strength of the external force and the initial particle trapping position. For intermediate Peclet numbers, the shapes of the collection efficiency curves were often found to be complex; for example, some conditions gave efficiency curves which showed local minima or maxima. Despite this complexity, a few simple observations were made: moving the particle trap away from the wafer (i.e., increasing $\tilde{h}$) always lowered the collection efficiency, as did reducing the attractive external force or increasing the repulsive force. Finally, calculations made for Reynolds numbers of 0 and 8 showed only modest variations in particle collection efficiency, suggesting that this parameter plays only a minor role over the range likely to be encountered in realistic processing environments.

Example calculations of collection efficiency were presented in dimensional terms for one representative set of process conditions (200 mm wafer, showerhead-to-wafer gap of 2.54 cm, mass flow rate of 1000 sccm argon, 1 torr, 300 K). In all cases, the efficiency curves exhibited a minimum near 0.1 μm, with increasing efficiency for both smaller and larger sizes. Trapping height was found to play a key role in deposition, and in all cases it was advantageous to manipulate the particle trap to a location as far from the wafer as possible. Trap manipulation can be accomplished in practice under some conditions. For example, in plasma processing, the trap location is determined by process parameters such as pressure, rf power, and flow rate; while these parameters may be fixed during etching by process requirements, they could be adjusted to manipulate the particle trap location just prior to plasma extinction. Similarly, trap position when particle nucleation is present could be controlled by pressure, wall temperature, flow rates, or chemistry selections.

At constant mass flow rate, collection efficiency was found to be independent of pressure for the low pressures and small particle sizes of interest. At constant pressure, collection efficiency was found to decrease significantly with increasing mass flow rate. Thus, from a particle transport point of view, reduction of particle-on-wafer counts could be obtained by increasing the mass flow rate in a process, assuming particle concentration varies inversely with mass flow rate rather than remaining independent. Another key result was that even modest temperature differences can lead to dramatic changes in collection efficiency due to the thermophoretic force. For example, deposition is nearly eliminated when the wafer is kept only 10-20°C warmer than the showerhead. On the other hand, the collection efficiency increases by an order of magnitude when the wafer temperature is decreased 10°C below the showerhead temperature.

Caution is suggested in implementing any of these strategy as the effect on the particle source term is not known.

2.8 INERTIA-ENHANCED DEPOSITION

The use of a showerhead restricts the area available to the flow inside the showerhead; consequently, the velocity of the gas inside the holes is much larger than the face velocity in the gap below. Particles originating upstream of the showerhead and suspended in the flow can be dramatically accelerated while passing through the showerhead, so that, at the exit of the showerhead, particle velocities much larger than the fluid face velocity are possible. Depending on conditions, particle acceleration by the showerhead can lead to inertia-enhanced particle deposition on the wafer below. Thus, a complete description of particle deposition on a wafer in a parallel-plate reactor must include a description of particle transport through the showerhead as well as an analysis of particle transport in the inter-plate region.

This section explores the role of inertia-enhanced deposition using the Lagrangian particle transport formulation given in Section 2.3. The strength of the Lagrangian formulation is in predicting particle transport resulting from the combination of applied external forces and particle inertia; the current implementation does not account for particle diffusion. The problem is separated into two domains in which particle and fluid transport are determined:(1) within a showerhead-hole and (2) between the showerhead and susceptor.

2.8.1 Particle Transport in the Showerhead Holes

Particles are assumed to be evenly distributed across the showerhead hole inlet and to enter with zero radial velocity and with an initial axial velocity equal to the face velocity U_0. The particle will immediately see a fluid velocity $U_{jet}(r)$, and will either be accelerated or deaccelerated by fluid drag depending on the magnitude of U_{jet}. Because of inertia, however, the particle will require a finite time to respond: in particular, the particle will accelerate to U_{jet} only if the showerhead hole is sufficiently long or the particle sufficiently small. In practice, the particle velocity at the exit of the showerhead, V_{po}, will fall somewhere between U_0 and U_{jet}; depending on the relative magnitudes of these two velocities, the showerhead thickness, and the particle relaxation time τ.

Assuming fully developed flow at the inlet and neglecting lift forces, the particle will remain at its initial radial position while in the tube; consequently, the axial fluid velocity driving the particle through the tube will also remain constant during the traverse. For plug flow, all particles will be accelerated by the same fluid velocity, $\overline{U}_{jet}$, independent of radial starting position. For fully developed parabolic flow, the local fluid velocity for each particle will depend on its starting position; particles near the wall will be slowed by drag, while particles near the centerline will be significantly accelerated. For parabolic flow, the assumption that the particles are evenly distributed (i.e., that the particle flux is constant) across the tube inlet is based on the assumption that the fluid entrance length is so short that particles do not have time to migrate radially as the flow is developing.[o]

The problem of a particle of given initial velocity experiencing a step-function change in the local fluid velocity is a classic problem. Although the problem could be solved in dimensional form, it is convenient for matching with the second part of this analysis if we solve the nondimensionalized particle equations of motion, Eqs. (2.19) and (2.20). As stated previously, we choose the mean chamber axial velocity U_0 as the characteristic velocity and the inter-plate spacing S as the characteristic length scale. External forces are assumed negligible compared to the large drag forces encountered in the showerhead holes. Given a particle of Stokes number St entering the hole with initial velocity U_0, and an axially constant fluid velocity $U_{jet}(r)$, Eqs. (2.19) and (2.20) can be solved analytically for the particle velocity at the showerhead exit, $z = L$:

$$\frac{L}{S} = \frac{U_{jet}}{U_0}\tilde{t} + St\left(\frac{U_{jet}}{U_0} - 1\right)\left(e^{-\frac{\tilde{t}}{St}} - 1\right) \tag{2.67}$$

$$\frac{V_{po}}{U_0} = \frac{U_{jet}}{U_0} - \left(\frac{U_{jet}}{U_0} - 1\right)e^{-\frac{\tilde{t}}{St}} \tag{2.68}$$

The solution procedure is as follows: (1) calculate U_{jet}/U_0, L/S, and St from process and particle parameters, (2) solve Eq. (2.67) for the dimensionless time $\tilde{t}$, and (3) use $\tilde{t}$ in Eq. (2.68) to solve for the dimensionless particle velocity at the showerhead exit. Equation (2.67) must be solved iteratively; for this purpose we use a Newton root-finding technique.

Some complexity has been added to Eqs. (2.67) and (2.68) by our use of one characteristic length and one characteristic velocity for both the showerhead and parallel-plate domains. An interesting result is found if L and U_{jet} (natural choices for characterizing transport through the showerhead) are substituted for S and U_0 as the characteristic length and velocity. With the appropriate redefinitions of the nondimensional terms, a set of equations similar to Equations (2.19) and (6.20) can be derived that depend on the jet Stokes number, which is related to our earlier definition by:

$$St_{jet} = \frac{\tau U_{jet}}{L} = St\frac{U_{jet}}{U_0}\frac{S}{L} \tag{2.69}$$

Using the same assumptions and initial conditions as above, we can derive a set of equations analogous to Eqs. (2.67) and (2.68) that are independent of L/S, i.e. $V_{po}/U_{jet} = f(U_{jet}/U_0,\ St_{jet})$. In addition, the functional dependence on the velocity ratio U_{jet}/U_0 is very weak (entering only as a result of the assumption

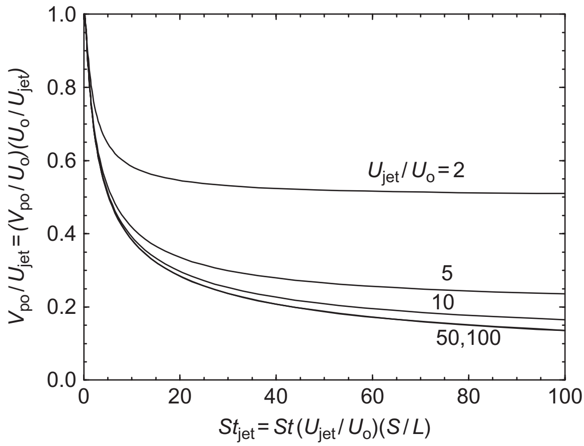

FIGURE 2.16 Acceleration of particles through the showerhead. Dimensionless velocity of particles exiting showerhead tubes, V_{po}/U_{jet}, as a function of jet Stokes number, St_{jet}, for a range of velocity ratios ($U_{jet}/U_0 = 2$, 5, 10, 50, and 100).

that the initial particle velocity is U_0), and vanishes for the limiting case where $U_{jet}/U_0 >> 1$ (which includes the case where the particle initial velocity is zero). In this limiting case, the ratio of the showerhead-exit velocity of the particle to U_{jet} depends only on the jet Stokes number. This result suggests plotting the results of Eqs. (2.67) and (2.68) as V_{po}/U_{jet} against St_{jet}—such as shown in Figure 2.16. As can be seen, the dimensionless particle velocity at the exit of the showerhead, V_{po}/U_{jet}, is reasonably insensitive to the velocity ratio when $U_{jet}/U_0 > 10$.

Thus, the analysis of particle transport in the showerhead domain is complete: given the three inputs $\overline{U}_{jet}/U_0$, L/S, and St (and particle initial radial position to determine $U_{jet}(r)$ for parabolic flow), we can calculate the dimensionless velocity V_{po}/U_0 at which the particle exits the showerhead. Interestingly, both the mean velocity ratio and the length ratio *depend only on reactor geometry;* process conditions such as temperature, pressure, or flow rate enter only through the Stokes number. Thus, for a given Stokes number, the extent of particle acceleration in the showerhead is entirely determined by hardware and is thus a characteristic of a specific tool design. Finally, in moving to the calculation of particle trajectories between the plates, the sign of the particle velocity V_{po}/U_{jet} must be switched to account for the different coordinate systems used in the two domains (see Figure 2.5b and the inset).

2.8.2 Particle Transport Between Parallel Plates

In this section, a Lagrangian formulation is used to calculate particle trajectories in the inter-plate region using both numerical FIDAP and analytical solutions of the flow field. Under the present assumptions, four dimensionless parameters uniquely determine particle transport in the inter-plate region: Re, St, V_p^t/U^0, and V_{po}/U_0 (a fifth—the interception parameter d_p/S—is neglected in this section). The first parameter, the Reynolds number, completely specifies the flow field for the infinite parallel-plate geometry—as demonstrated in the analytical low-Re approximation to the flow field given in Eq. (2.46). For a finite-plate geometry, the aspect ratio is also needed to specify the flow. The second parameter, the particle Stokes number, is used in this work as a dimensionless particle diameter—as suggested by the free molecule limit Eq. (2.23). The influence of external forces enters through the third parameter, the dimensionless particle drift velocity, which parameterizes the forces via the z-component of the net drift velocity, V_p^t. The fourth parameter, the dimensionless particle velocity at the showerhead exit, is determined by the strength of the showerhead acceleration effect as described in Section 2.8.1. Initially, however, V_{po}/U_0 will be taken as an independent variable; later, we discuss the coupling of the showerhead and parallel-plate domains.

The effect of the these dimensionless parameters is shown in Figure 2.17, where particle collection efficiency is plotted against Stokes number for $Re = 8$. For Figure 2.17a, an initial dimensionless particle velocity $V_{po}/U_0 = -1$ is assumed (no particle showerhead acceleration), while in Figure 2.17b the initial velocity is taken as -100 (substantial showerhead acceleration characteristic of commercial reactors). In each plot, the influence of external forces is explored by varying the drift velocity: curves for $V_p^t/U^0 = -0.5, -0.1, -0.01, 0, 0.1$, and 1.0 are shown. Negative values of the drift velocity correspond to an external force directed toward the wafer (attractive, enhancing deposition), while positive values correspond to an external force directed away from the wafer (repulsive, inhibiting deposition). Several important features of these plots will now be explored.

First, inertial effects lead to particle deposition even in the absence of external forces as shown by the curves for $V_p^t/U^0 = 0$ in Figure 2.17. In this case there is a critical Stokes number, St_{crit}, below which no deposition occurs. At St_{crit} there is a sharp jump in efficiency, which then increases toward unity with increasing St. The jump is steeper, higher (approaching unit collection efficiency) and occurs for a much smaller St_{crit} in the case with substantial showerhead acceleration than when the particles enter with the fluid face velocity. This effect is discussed in greater detail below. As seen in Figure 2.17, particle inertia can also lead to deposition even when an external force is pushing particles away ($V_p^t/U^0 > 0$). The extent of external force "protection" is significantly reduced for large initial velocities: compare the $V_{po}/U_0 = 0$ and 1 curves in Figure 2.17a and b.

Second, when an external force is directed toward the wafer ($V_p^t/U^0 < 0$), particle deposition occurs at all values of the Stokes number. In the small-St limit (negligible particle inertia), the collection efficiency should tend to Eq. (2.48) for external forces that can be described by a potential; this trend is clearly evident in Figure 2.17. For example, for $V_p^t/U^0 = -0.5$, Eq. (2.48)

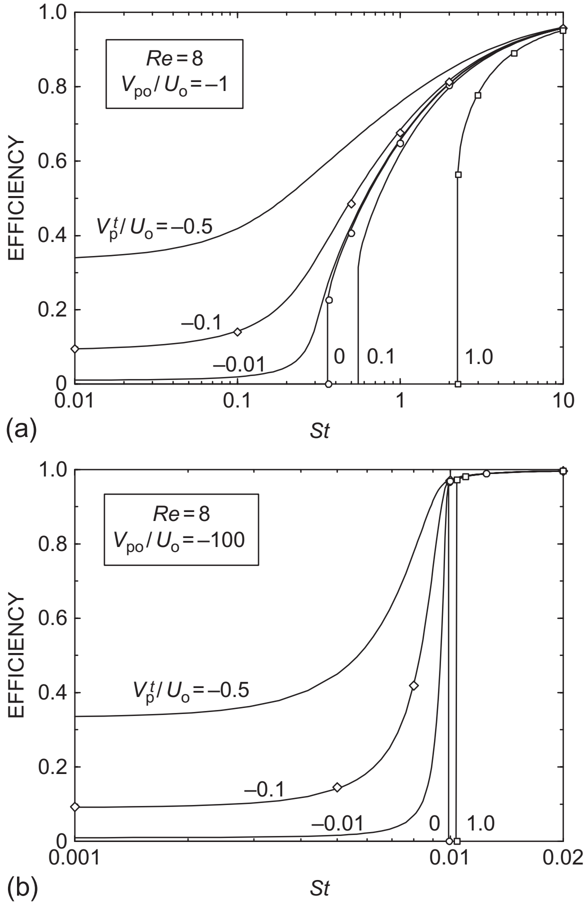

FIGURE 2.17 Efficiency vs. Stokes number for various deposition velocities for $Re = 8$. Solid lines are calculated using an analytical flow field and a Runga-Kutta integrator, while the symbols are calculated using numerical flow solutions and the FIDAP particle tracking post-processing routines. (a) Particle dimensionless initial velocity of -1 and (b) particle dimensionless initial velocity of -100.

predicts an efficiency of 1/3 which is the observed asymptote of the appropriate curves in Figure 2.17a and b. The presence of an attractive external force smooths the shape of the efficiency curves for large attractive forces; however, when the magnitude of the attractive force is small (e.g., $V_p^t/U^0 = -0.1$), the efficiency still exhibits a sharp increase in the neighborhood of St_{crit} (from the no-force case). The rise in efficiency near St_{crit} is much steeper for high initial particle velocities.

Finally, in the large-*St* limit, the collection efficiency must approach unity for all cases. That is, for large enough Stokes numbers, particle inertia leads to straight trajectories and complete deposition independent of the details of the flow field, the external forces, or particle initial velocity (assuming it is not zero). This limit is approached in all of the calculations shown in Figure 2.17.

2.8.2.1 Asymptotic Limit of Critical Stokes Number

An interesting result is suggested by Figure 2.17b for the case of no external force: as the particle initial velocity becomes large, the collection efficiency tends toward a step function which jumps from zero to unity at a critical Stokes value equal to the inverse of the dimensionless initial particle velocity. To confirm this result, a series of calculations were made to explore the dependence of the critical Stokes number on initial particle velocity in the absence of an external force. These results are shown in Figure 2.18, where St_{crit} is plotted as a function of V_{po}/U_0 for fluid Reynolds numbers of 0, 4, and 8. For a given value of V_{po}/U_0, particles with $St < St_{\mathrm{crit}}$ (below the line) will exit the reactor, while particles with $St > St_{\mathrm{crit}}$ (above the line) will impact the reactor. The effect of Reynolds number is negligible for large values of V_{po}/U_0; in fact, for $V_{\mathrm{po}}/U_0 > 10$ the three Re curves approach the same asymptotic limit. For large values of V_{po}/U_0 particle inertia dominates deposition and the details of the flow field become unimportant. Inspection of the large initial-velocity asymptotic limit reveals the following relationship:

$$St_{\mathrm{crit}} \rightarrow \frac{U_{\mathrm{o}}}{V_{\mathrm{po}}} \tag{2.70}$$

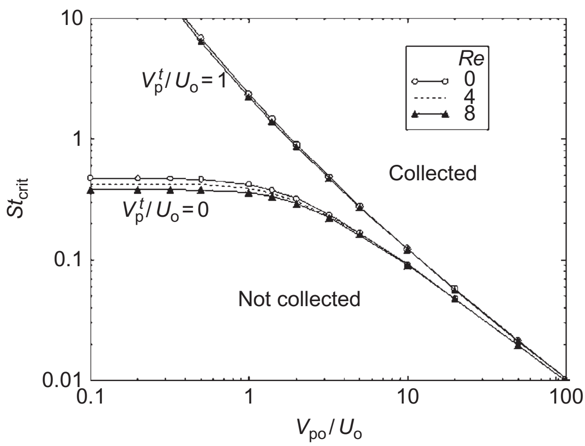

FIGURE 2.18 Critical Stokes number vs. particle dimensionless inlet velocity. Values of the critical Stokes number were calculated using the analytical approximation to the flow field for Reynolds numbers of 0, 4, and 8. One set of curves applies for no external force ($V_{\mathrm{p}}^t/U_0=0$), the other set applies for a strong force resisting deposition ($V_{\mathrm{p}}^t/U_0=1$).

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A simple explanation of this limit is readily illustrated by rearranging Eq. (2.70) to give $St_{\text{crit}} V_{\text{po}}/U_0 = \tau\, V_{\text{po}}/S = 1$, which states that impaction occurs when the particle stopping distance based on its *initial* velocity V_{po} equals the showerhead-to-wafer gap.[p]

Figure 2.18 also shows the variation of the critical Stokes number when a large external force opposing deposition is applied ($V_p^t U_0 = 1$). Although the value of St_{crit} is greatly increased for small V_{po}/U_0 (compared to the case with no external force acting), all curves approach the same asymptote—Eq. (2.70)—in the limit of very large initial particle velocity. The St_{crit} value for a large external force was found to be ~4% higher than without an external force for $V_{\text{po}}/U_0 = 100$. As noted above, the influence of Reynolds number on St_{crit} is greatly reduced when a repulsive force is acting.

Thus, for no or repulsive external forces, a great simplification results for large values of the initial particle velocity (say for $V_{\text{po}}/U_0 > 100$): the collection efficiency can be closely approximated by a step function (from zero to unity) at a critical Stokes number calculated by Eq. (2.70). When an attractive force is present, the concept of a critical Stokes number breaks down, as there is some deposition at all Stokes numbers. Even in this case, however, a sharp increase in efficiency near St_{crit} is still seen (such as shown in Figure 2.17b).

2.8.3 Coupled Transport—Nondimensional Results

In this section, showerhead-enhanced inertial deposition is explored by coupling the transport of particles through the showerhead and in the inter-plate region. The procedure is as follows: (1) for given values of U_{jet}/U_0, L/S, and St, calculate the dimensionless velocity of the particle exiting the showerhead, V_{po}/U_0, using Eqs. (2.67) and (2.68); and then (2) using V_{po}/U_0 as the initial particle velocity, and the parameters Re, St, and V_p^t/U_0, integrate the particle trajectory between the plates to determine the particle collection efficiency. Thus, the coupled particle transport problem (for an infinite parallel-plate geometry and under the present assumptions) is completely specified by five independent dimensionless parameters (note that V_{po}/U_0 is dependent). Efficiency results from these coupled calculations should look qualitatively like those shown in Figure 2.17, although some variations are expected as the initial particle velocity is no longer fixed but depends on the degree of particle acceleration through the showerhead.

It is valuable at this point to clarify the use of the jet to face-velocity parameter U_{jet}/U_0, which is the local fluid velocity that a particle experiences while passing through a showerhead hole. For the assumption of plug flow through the showerhead, $U_{\text{jet}}/U_0 = \overline{U}_{\text{jet}}/U_0 = A_{\text{showerhead}}/\Sigma A_{\text{jet}}$ where $\overline{U}_{\text{jet}}$ is the mean velocity in the tube. In the plug-flow case, U_{jet}/U_0 must always be larger than unity and is constant across each showerhead tube cross-section. In commercial reactors, values of $\overline{U}_{\text{jet}}/U_0$ are seldom less than 20, and can range up to several hundred. The other limit considered in this work is parabolic flow through the showerhead holes. In this case, U_{jet}/U_0 is function of both the area ratio and the radial starting position of the particle in the showerhead tube. For example, a particle starting on the tube centerline would experience a fluid velocity twice the mean, so that $U_{jet}(r=0)/U_0 = 2\bar{U}_{jet}/U_0$. Because the fluid velocity must vanish at the tube wall, jet to face-velocity ratios less than one are possible for the parabolic case for particles starting near the wall. In the following, results are parameterized with the most general form U_{jet}/U_0.

2.8.3.1 Critical Stokes Numbers

One set of coupled efficiency calculations is shown in Figure 2.19, which plots efficiency versus Stokes number for jet to face-velocity ratios of 0.1, 1, 10, 100, and 1000. For these calculations, the showerhead thickness was assumed equal to the plate gap ($L/S=1$) for the case of $Re=0$ and no external forces acting ($V_p^t/U_0=0$). The critical Stokes number (the smallest St for which collection occurs) is found to decrease with increasing values of U_{jet}/U_0. This result is not surprising: as U_{jet}/U_0 increases the particle velocity at the showerhead exit (V_{po}/U_0) must also increase, and we have shown in Section 2.8.2 that increasing values of V_{po}/U_0 lead to smaller values for St_{crit} (see Figure 2.18). In particular, we have shown in the limit of large V_{po}/U_0 that St_{crit} $U_0/\rightarrow V_{po}$. It is interesting to note that, for coupled transport, the large U_{jet}/U_0 limit of St_{crit} is *not* U_0/U_{jet}, but a slightly higher value (e.g., for $U_{jet}/U_0=100$, $St_{crit}=0.01237$). This difference is explained by the fact that, because of inertia, the particle cannot

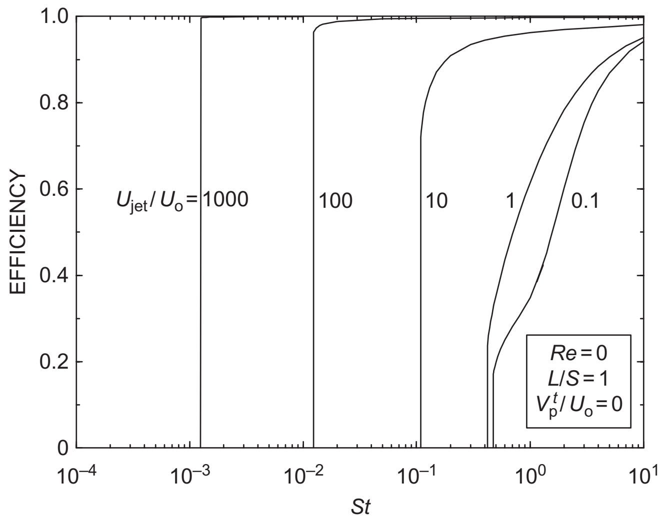

FIGURE 2.19 Effect of jet to face-velocity ratio. Efficiency vs. Stokes number for $U_{jet}/U_0=1$, 10, 100, and 1000 including coupling between showerhead and inter-plate transport (for this calculation $L/S=1$, $Re=0$, $V_p^t/U_0=0$, and $r_p/S=1\times10^{-7}$).

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accelerate to the jet velocity before exiting the showerhead (i.e., V_{po}/U_0 U_{jet}/U_0); consequently, a larger Stokes number is required to initiate deposition. In the slow-jet limit, say $U_{jet}/U_0 < 1$, the critical Stokes number becomes less sensitive to the particle inlet velocity, although the shapes of the efficiency curves can be quite different (e.g., compare the curves for $U_{jet}/U_0 = 0.1$ and 1). Here, the details of the efficiency curve result from a complicated interplay among the parameters. Overall, reducing the value of U_{jet}/U_0 (more and/or larger showerhead holes) increases St_{crit}—with the favorable result of increasing the minimum size for which inertial effects lead to particle deposition on the wafer.

The showerhead thickness also plays a role in determining the magnitude of the critical Stokes number. The effect of showerhead thickness is presented in Figure 2.20, in which collection efficiency is plotted against St for L/S =0.1, 0.5, 1 and 2 (for $U_{jet}/U_0 = 100$, $Re = 0$, and $V_p^t/U_0 = 0$). For large L/S values, there is sufficient time in the showerhead for the particle to accelerate to the jet velocity (V_{po}/U_0 U_{jet}/U_0), which by Eq. (2.70) gives the asymptotic limit $St_{crit}U_0/U_{jet}$. As seen in Figure 2.20, this limit is approached for $L/S > 2$. For very thin showerheads, the particles spend only a short time in the showerhead, and will exit the showerhead with a velocity much less than the jet velocity. In this case, larger Stokes numbers are needed to initiate deposition—as seen in Figure 2.20 for the curves with $L/S = 0.1$ and 0.5. Based on these results, St_{crit} can be increased (and inertial deposition reduced) by reducing the dimensionless showerhead thickness L/S.

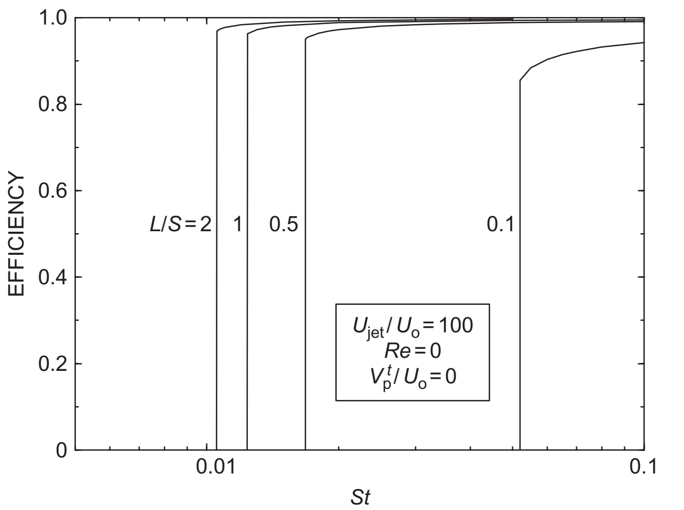

FIGURE 2.20 Effect of showerhead thickness. Efficiency vs. Stokes number for $L/S = 0.1$, 0.5, 1, and 2 including coupling between showerhead and inter-plate transport (for this calculation $U_{jet}/U_0 = 100$, $Re = 0$, $V_p^t/U_0 = 0$, and $r_p/S = 1 \times 10^{-7}$).

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2.8.3.2 Grand Design Curves

If the effect of a repulsive force is neglected, then the value of St_{crit} (for a given flow field) is determined solely by the values of U_{jet}/U_0 and L/S. Interestingly, both of these parameters are geometrical in nature. The geometric interpretation of L/S is obvious (the ratio of showerhead thickness to inter-plate gap), while that for U_{jet}/U_0 requires some explanation. Under the assumption of plug flow within the showerhead holes, it has already been shown that $U_{jet}/U_0 = \overline{U}_{jet}/U_0 = A_{showerhead}/\Sigma A_{jet}$ where $\overline{U}_{jet}$ is the mean velocity in the tube. Thus, in the plug-flow limit the velocity ratio is completely specified by the number and size of the showerhead holes and by the diameter of the showerhead—purely geometric properties of the showerhead. Under the assumption of parabolic flow, the radial starting position of the particle within the showerhead hole must also be considered, but this is another geometrical parameter. Thus, for a specific flow field and neglecting external forces, the critical Stokes number is uniquely specified by chamber and showerhead geometry (and possibly an assumed particle starting position), and is independent of process parameters (e.g., gas temperature, pressure, or flow rate).

This simplification leads to the idea of the grand design curves, which give critical Stokes number as a function of the velocity ratio U_{jet}/U_0 for various dimensionless showerhead thicknesses L/S. An example of a grand design curve over a wide range of these two parameters is shown in Figure 2.21 for $Re = 0$ and $V_p^t/U_0 = 0$. The qualitative trends are consistent with earlier discussion: the critical Stokes number decreases with increasing values of U_{jet}/U_0 and L/S. The curves for all values of L/S intersect at $U_{jet}/U_0 = 1$, which corresponds to the

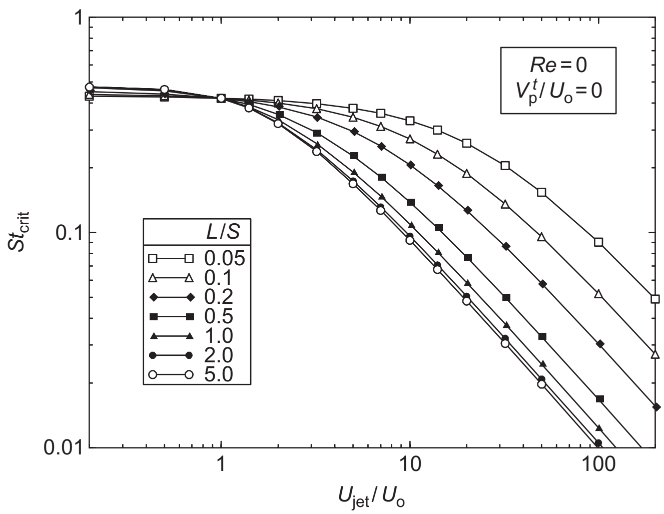

FIGURE 2.21 Grand design curve for estimating the critical Stokes number based on showerhead parameters (for this calculation $Re = 0$, $V_p^t/U_0 = 0$, and $r_p/S = 1 \times 10^{-7}$).

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case for which the particle enters (and exits) the showerhead at the same velocity. Variations in St_{crit} with L/S become small for dimensionless showerhead thicknesses larger than about two; in this case, the particle has had sufficient time to accelerate to near the gas velocity, so that making the showerhead longer has little effect. For velocity ratios less than one, the critical Stokes number is essentially independent of velocity ratio and the dimensionless showerhead thickness. For large velocity ratios, the curves in Figure 2.21 for different L/S values become parallel with a slope of unity—suggesting that in this limit St_{crit} becomes proportional to U_0/U_{jet}. As shown, the value of U_{jet}/U_0 at which this asymptotic limit is reached depends on L/S: for thick showerheads the linear limit is achieved at much smaller velocity ratios than for thin showerheads.

For design applications, the grand design curves are used with the parameters U_{jet}/U_0 and L/S to find the critical Stokes number for the proposed reactor geometry. To minimize particle deposition on the wafer, it is desirable to choose parameters that give as large a critical Stokes number as possible, as increasing St_{crit} increases the minimum size at which inertial deposition begins. Based on Figure 2.21, larger values of St_{crit} are obtained by decreasing U_{jet}/U_0 (use more and/or larger showerhead holes) or by decreasing L/S (use a thin showerhead or a large inter-plate gap). Introducing a force that opposes deposition (such as by heating the wafer relative to the showerhead) will always help, but as discussed above, the effect is fairly small under realistic conditions. Once St_{crit} has been determined, the corresponding particle critical size, $d_{p,crit}$, can be found from Eq. (2.21). Note that although the critical Stokes number depends only on geometric parameters, the critical particle diameter depends on geometric, process, and particle parameters. Thus, particle density and gas pressure, temperature, type, and flow rate all play a role in determining $d_{p,crit}$. The effects of process parameters on showerhead-enhanced inertial deposition are discussed in a later section.

2.8.3.3 External Forces

Although particle deposition only occurs for $St > St_{crit}$ for repulsive or zero external forces, inertial deposition will always take place when the net external forces are attractive. To demonstrate this behavior, particle collection efficiencies calculated using fully coupled particle transport (i.e., including showerhead acceleration) are shown for various values of the external force in Figure 2.22 (for $Re = 0$, $U_{jet}/U_0 = 100$, $L/S = 1$). The results are very similar to those in Figure 2.17b, which gives efficiency for a fixed particle inlet velocity ($V_{po}/U_0 = 100$) instead of the present case where the showerhead exit velocity is calculated based on showerhead parameters.[q] The large and small Stokes limits are the same: for small St inertial effects vanish and the efficiency must tend to Eq. (2.48), while for large St inertia dominates and efficiency must tend to unity. At intermediate values of St there are some differences. For example, it is seen that the critical Stokes number for the coupled analysis is slightly larger

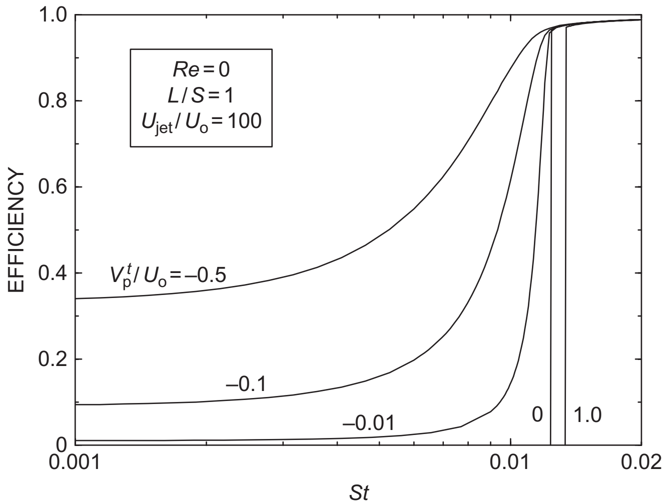

FIGURE 2.22 Efficiency vs. Stokes number for various deposition velocities. Collection efficiencies for fully coupled particle transport for V_p^t/U_0 values of -0.5. -0.1, 0, and 1 (for this calculation $L/S = 1$, $Re = 0$, $U_{jet}/U_0 = 100$, and $r_p/S = 1 \times 10^{-7}$).

than for the case where V_{po}/U_0 is held constant; as discussed above, for a finite-length showerhead the particle velocity at the showerhead exit must be slightly less than U_{jet}/U_0, so that a larger Stokes number is needed to initiate inertial deposition. Note that the case of constant V_p^t/U_0 (i.e., V_p^t/U_0 is independent of particle size or Stokes number) corresponds to the physically meaningful situation in which a thermophoretic force is acting (as the thermophoretic drift velocity is independent of particle diameter). For other external forces—such as gravity—the net drift velocity will not be a constant but will vary with particle diameter (and hence with St).

2.8.3.4 Parabolic Profile

For fully developed parabolic flow in the showerhead hole, the gas velocity varies with radial position within the hole as given by Eq. (2.45); in this case the local velocity experienced by a particle in a showerhead hole depends on its radial starting position and the mean velocity $\overline{U}_{jet}$ [as given by Eq. (2.44)]. For example, a particle starting on the hole centerline will experience the highest local gas velocity ($2\overline{U}_{jet}$), while ones starting near the hole wall will experience much lower velocities. As the amount of acceleration, the particle experiences within the showerhead depends on the local gas velocity, the particle collection efficiency must vary with particle radial position within the showerhead hole. Thus, the calculation of the net collection efficiency for the parabolic flow case requires integrating the local efficiency radially across the showerhead hole. Assuming that the particles are uniformly distributed across the showerhead hole (i.e., that the flux of particles across the tube cross-section is constant), the net efficiency is:

$$\eta_{\mathrm{net}}(St) = \frac{2}{R_{\mathrm{jet}}^2}\int_0^{R_{\mathrm{jet}}} \eta\left[U_{\mathrm{jet}}(r), St\right] r\,\mathrm{d}r \tag{2.71}$$

The numerical integration of Eq. (2.71) is computationally expensive, as each evaluation of the integrand $\eta[U_{\mathrm{jet}}(r), St]$ requires a coupled calculation of the particle acceleration through the showerhead [with local gas velocity $U_{\mathrm{jet}}(r)$] along with the corresponding numerical integration of the particle trajectory between the two plates. The integration of Eq. (2.71) is further complicated by the fact that, in some cases, efficiency can change significantly with small changes in St or $U_{\mathrm{jet}}(r)$. In the present work, an adaptive Gauss integration scheme with automatic error control has been used to evaluate Eq. (2.71).

An example of a net efficiency curve for a parabolic velocity profile is shown in Figure 2.23 for the case where there is no external force ($Re = 0$, $\overline{U}_{\mathrm{jet}}/U_0 = 100$, $L/S = 1$, and $V_p^t/U_0 = 0$). For comparison, efficiency curves are also shown for plug flow (where all the particles experience the mean velocity) and for the hypothetical case where all of the particles experience the centerline velocity (essentially a plug flow moving at twice the mean velocity). These three curves show three interesting effects. First, the critical Stokes

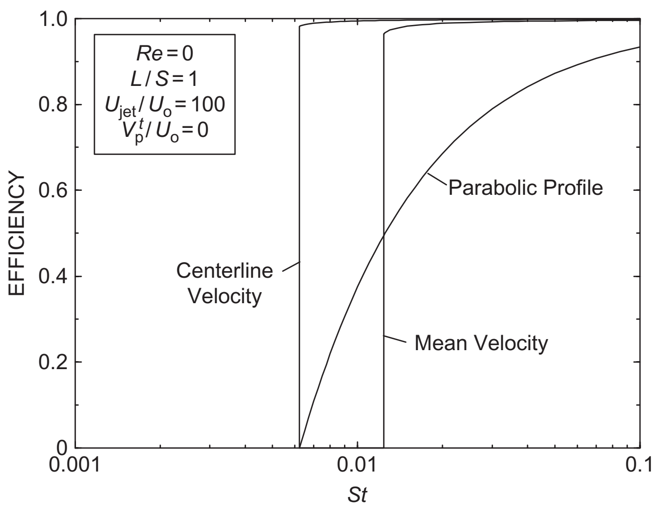

FIGURE 2.23 Efficiency vs. Stokes number for parabolic showerhead profile (no external force). Collection efficiencies for fully coupled particle transport for particles experiencing the showerhead hole centerline and mean velocities (plug flow assumption), and integrated over the parabolic velocity profile in the showerhead holes in the absence of external forces ($L/S = 1$, $Re = 0$, $\overline{U}_{\mathrm{jet}}/U_0 = 100$, and $r_p/S = 1 \times 10^{-7}$).

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number for the centerline case is approximately one half that for the plug flow case; this result is explained by the fact that the critical Stokes number is inversely proportional to jet velocity in the limit of large values of U_{jet}/U_0 (i.e., doubling the local showerhead gas velocity halves St_{crit}).[r] Second, the efficiency curve for the parabolic case approaches unity much slower than either of the plug flow curves. This is expected, as particles located near the tube wall experience very low local velocities; for some region very close to the wall, particles actually deaccelerate while passing through the showerhead. Note that although this slow approach to complete collection is favorable from a defect reduction point of view, it is not of practical significance because in this large Stokes regime the majority of particles entrained in the flow would still be deposited.

Third, the critical Stokes number for the parabolic case is the same as for the centerline case. The smallest particles deposited are those experiencing the highest velocity in the showerhead hole; thus, for parabolic flow inertia-enhanced deposition begins with those particles moving along the showerhead-hole centerline. The slope of the parabolic case is not as steep as for the centerline case because few particles are contained in the small-area region near the centerline, whereas in the centerline case we have assumed all of the particles are moving at the centerline velocity. The centerline case therefore serves as a limit of the smallest Stokes value for which inertial-enhancement to deposition becomes important in parabolic flow. In fact, the centerline case serves as a lower limit for all laminar flow conditions in the tubes, since even for developing flow the maximum velocity in the tube will always be less than $2\overline{U}_{jet}$. Thus, the most conservative practice for predicting the effects of inertia-enhanced deposition in real reactors is to use $U_{jet}/U_0 = 2\overline{U}_{jet}/U_0$ as the characteristic velocity ratio in the grand design curves.

The present results show that, for parabolic flow in the showerhead, the best practice for determining net efficiency is to perform the full integration of Equation (6.71). However, since this calculation can be computationally expensive, the following approximations are suggested: for attractive external forces (say $V_p^t/U_0 < -0.01$) use the mean velocity approximation, otherwise (say $V_p^t/U_0 > 0$) use the more conservative centerline approximation.

2.8.4 Coupled Transport—Dimensional Results

The two approaches to reduce inertia-enhanced particle deposition are[s]: (1) design equipment with as large a value of St_{crit} as possible and (2) select process conditions that give as high a value of $d_{p,crit}$ as possible. Based on our previous analysis, the only three ways to increase St_{crit} is to design for minimum $\overline{U}_{jet}/U_0$ and/or L/S, and to apply an external force that opposes deposition. To minimize $\overline{U}_{jet}/U_0$, very porous showerhead designs are needed to reduce the constriction of the flow (decrease the area ratio given by Eq. (2.44) by either increasing the number or the size of holes). The ratio L/S can be reduced by reducing the showerhead thickness or by increasing the showerhead-to-wafer gap. Intuitively, a short showerhead thickness reduces the time available to accelerate the particle, and a large showerhead-to-wafer gap provides the particle more opportunity to slow down. Finally, an opposing external force could be used to inhibit inertial deposition such as by keeping the wafer warmer than the showerhead to take advantage of thermophoresis but remembering that the opposing force typically had a fairly weak effect on reducing inertia-enhanced deposition.

Once a hardware design is fixed, St_{crit} is fixed, but it is still possible to minimize inertial deposition by selecting process conditions that give as high a value of $d_{p,crit}$ as possible. Based on the free molecule definition of Stokes number in Eq. (2.23), we can see that low face velocities U_0 and large showerhead-to-wafer gaps S are preferred. Mean gas velocities can be reduced by reducing mass flow rates (at constant pressure) or by operating at higher pressures (for a fixed mass flow rate). As seen in Eq. (2.23), operating at higher pressures also directly increases $d_{p,crit}$. Thus, for a constant mass flow rate, a twofold increase in pressure produces a fourfold increase in $d_{p,crit}$ (one factor of two directly from the pressure reduction, and an additional factor of two from lowering the face velocity).[t] Recall that Eq. (2.23) strictly applies to particles in the free molecular limit (small sizes and/or low pressures) which is a reasonable assumption for the pressure and particle size regimes in typical semiconductor manufacturing reactors.

An example of particle collection efficiency as a function of particle size is shown in Figure 2.24 for a hypothetical 200 mm (diameter) reactor

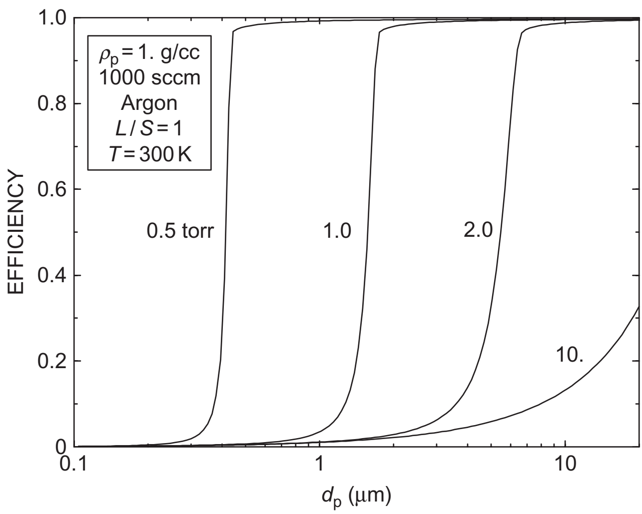

FIGURE 2.24 Efficiency vs. particle diameter and pressure (isothermal case). Collection efficiencies for fully coupled particle transport assuming plug flow through the showerhead hole for reactor pressures of 0.5, 1.0, 2.0, and 1 torr ($L/S = 2.54$ cm, $\overline{U}_{jet}/U_0 = 99.2$, $Q = 1000$ sccm argon, $T = 300$ K, $Re = 0.98$, and particle density $= 1$ g/cm^3).

characterized by: an argon flow rate of 1000 sccm (standard cubic centimeters per minute) through a showerhead 2.54 cm thick with 1000 holes of diameter 0.0635 cm with a showerhead-to-wafer gap of 2.54 cm. The flow is assumed isothermal and a particle density of 1 g/cm^3 is used. These reasonable physical parameters give the dimensionless quantities $L/S=1$, $\overline{U}_{jet}/U_0 = 99.2$, and $Re=0.984$; the dimensionless drift velocity varies with size according to the expression for gravitational settling velocity—Eq. (2.26). Efficiency curves are shown for reactor pressures (i.e., pressure between the two plates) of 0.5, 1.0, 2.0, and 10. torr. For pressures less than 2 torr, inertial-enhancement to deposition is clearly evident by the abrupt jump in efficiency from nearly zero to unity in the vicinity of a critical size. As seen, this critical size is a strong function of pressure, with an approximately fourfold decrease in the critical size for a twofold decrease in chamber pressure. For chamber pressures above about 10 torr inertial effects no longer contribute to deposition; the increase in efficiency for increasing particle size seen in Figure 2.24 for the $P=10$ torr case results because the gravitational drift velocity increases with size.

The same reactor geometry and process conditions have been used to calculate the effect of thermophoresis on collection efficiency as shown in Figure 2.25. The showerhead temperature has been held constant at 300 K while the wafer (lower plate) temperature is made colder (280 K), isothermal (300 K), or hotter (320 K). Fairly modest temperature differences have been used in accordance with our isothermal (constant gas properties) assumption for flow calculations, but that small temperature differences are allowed to drive particle

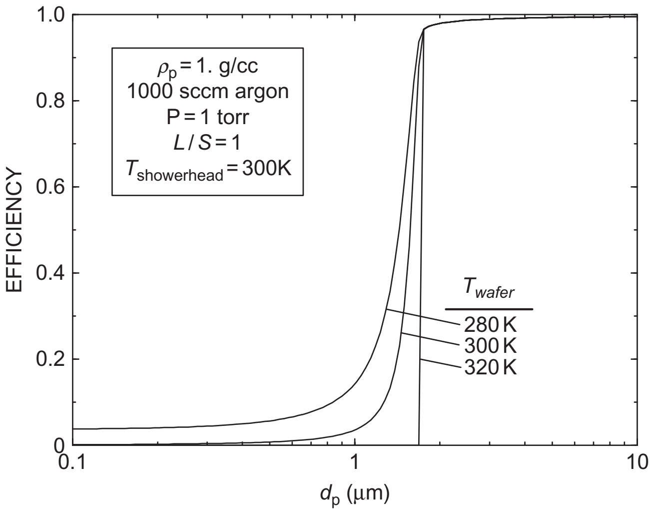

FIGURE 2.25 Efficiency vs. particle diameter and wafer temperature (with thermophoresis). Collection efficiencies for fully coupled particle transport assuming plug flow through the showerhead hole for wafer temperatures of 280, 300, and 320 K ($L/S=2.54$ cm, $\overline{U}_{jet}/U_0=99.2$, $Q=1000$ sccm argon, $P=1$ torr, $T_{showerhead}=300$ K, $Re=0.98$, and particle density $=1$ g/cm^3).

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thermophoresis. The isothermal case shows that, for small particle sizes, deposition decreases with decreasing size because the gravitational drift velocity is decreasing. When the wafer temperature is less than the showerhead temperature, thermophoresis acts to increase deposition compared to the isothermal case. Although the differences become small for sizes larger than $d_{p,crit}$ (about 1.5 μm), the thermophoretic contribution is clear in the small-particle limit where the efficiency approaches a constant.[u] Heating the wafer relative to the showerhead eliminates all small-particle deposition as the thermophoretic resistive force overwhelms the attractive gravitational force. The critical diameter at which inertial effects dominate is shifted to a slightly larger size than for the isothermal case, although the shift is fairly small since the effect of an external force on St_{crit} is fairly weak. By using thermophoretic protection and by designing a reactor with a sufficiently large $d_{p,crit}$ particle deposition can be mitigated over the particle size range of interest.

The two previous examples highlight two important points: (1) that inertia-enhanced deposition becomes dramatically more important at low pressure and (2) that thermophoretic protection coupled with careful control of inertial effects can significantly reduce particle deposition. Dimensional calculations could also be used to demonstrate all of the effects reported in Section 2.8.3, however, because of the wide variability in reactor geometry and particle and process parameters, only these two examples are presented.

2.9 CHAPTER SUMMARY AND PRACTICAL GUIDELINES

This chapter reviewed the basic phenomena controlling particle transport and the underlying general equations with an emphasis on conditions encountered in semiconductor process tools (i.e., subatmospheric pressures and submicron particles). The discussion included expressions for the following particle forces: fluid drag, gravity, thermophoresis, and electrophoresis. The concepts of particle drift velocity and stopping distance were introduced, and issues of continuum vs. free molecular particle transport were outlined. Particle concentrations were assumed to be low enough to allow a dilute approximation, for which the coupling between the fluid and particle phases is one-way. In this case, the fluid/thermal transport equations can be solved either analytically or numerically neglecting the particle phase; the resulting velocity and temperature fields were then used as input for the particle transport calculations. Isothermal flow was assumed, although small temperature differences were allowed to drive particle thermophoresis; both analytical and numerical solutions of the flow field representative of a parallel plate geometry were presented. Particle collection efficiency was defined as the fraction of particles present in the inter-plate region of the reactor that deposit on the wafer. Particles were presumed to either enter the reactor through the showerhead (uniformly spread between $r = 0$ and R_W), or to originate in a plane parallel to the wafer. The latter case corresponds to particles being released from a plasma trap upon plasma extinction or being formed in a nucleation process; in this case, the particles are initially assumed to be uniformly spread radially between $r = 0$ and R_W at some distance h from the wafer. Analytical expressions for collection efficiency were presented for the limiting case where external forces control deposition (i.e., neglecting particle diffusion and inertia).

Particle transport in the parallel-plate geometry was predicted using both the Lagrangian approach (where individual particle trajectories are calculated) and the Eulerian approach (where the particles are modeled as a cloud). The Eulerian formulation yielded an analytical, integral description of particle deposition for the case where the flow field between the plates can be approximated analytically. The strength of the Eulerian formulation is in predicting particle transport resulting from the combination of applied external forces (including the fluid drag force) and the chaotic effect of particle Brownian motion (i.e., particle diffusion), although the current implementation cannot account for particle inertia. In particular, the Eulerian formulation cannot accommodate particle acceleration effects within the showerhead, and is therefore restricted to particle transport in the inter-plate region.

The need to properly account for diffusion-enhanced particle deposition becomes increasingly important as the semiconductor industry moves toward smaller feature sizes and becomes concerned with smaller-sized particles. Based on the Eulerian analysis, the following guidelines are intended to help tool operators and designers reduce particle deposition when diffusional effects are important:

- Keep traps as far from the wafer as possible.
- Take advantage of repulsive forces, such as thermophoretic protection gained by keeping the wafer warmer than the showerhead.
- Reduce attractive forces.
- For a specific pressure, use as high a mass flow rate as possible.

The strength of the Lagrangian formulation is in predicting particle transport resulting from the combination of applied external forces and particle inertia, although the current implementation cannot account for particle diffusion. It is the Lagrangian formulation that can properly account for inertia-enhanced deposition resulting from particle acceleration in the showerhead. The problem was treated in two steps, in which both particle and fluid transport were determined: (1) within a showerhead hole and (2) between the showerhead and the susceptor. For fluid and particle transport in the showerhead, approximate analytical expressions were derived based on a few assumptions. The output of this first step was the particle velocity at the exit of the showerhead, as a function of showerhead geometry, flow rate, and gas and particle properties. The particle showerhead-exit velocity was next used as an initial condition required for particle transport between the plates. The output of the second step was a prediction of particle collection efficiency by the susceptor (wafer), as a function of showerhead-exit particle velocity, the plate separation, flow rate, and gas and particle properties. Based on the Lagrangian analysis, two approaches were identified to help tool operators and designers reduce particle deposition when inertial effects are important: (1) design equipment with as large a value of St_{crit} as possible and (2) select process conditions that give as high a value of $d_{p,crit}$ as possible. Based on this analysis, St_{crit} is determined primarily by reactor and showerhead geometry. Only three methods were identified for increasing St_{crit}:

- Decrease the showerhead velocity ratio $\overline{U}_{jet}/U_0$ by increasing the number of holes and/or enlarging the size of the showerhead holes.
- Decrease the showerhead thickness ratio L/S by making the showerhead very thin or the inter-plate gap large.
- Apply an external force that opposes particle deposition (such as keeping the wafer warmer than the adjacent gas).

Given a specific hardware design (and corresponding St_{crit}), inertial deposition can be reduced by selecting process conditions that give as high a value of $d_{p,crit}$ as possible. Based on the free molecule limit of Stokes number, the following general guidelines are offered to increase the critical diameter (for a given St_{crit}):

- Increase the gap between the showerhead and wafer
- Use low mass flowrates
- Raise chamber pressure
- Use a high molecular weight gas

The previous recommendations specifically pertain to reducing particle deposition given an assumed dominant deposition mechanism; note that one set of guidelines (e.g., for inertia) may conflict with those intended to reduce deposition by other mechanisms (e.g., gravity or diffusion). In order to reduce particle deposition in real tools, it is up to equipment designers/operators to first identify the dominant deposition mechanism so that an effective improvement strategy can be identified. Note that the guidelines given above are not intended to replace detailed calculations (using the proper analysis with the actual process conditions), but to provide the user with a general feel for inherently-clean practices. In addition, equipment designers should be aware that while these recommendations should improve particle performance, the effect of any changes on process performance must also be investigated.

NOTES

a The flow field entraining the particle can also be either molecular or continuum in nature. In this case, discrimination between the two flow regimes is made by comparing some characteristic length associated with the reactor geometry to the gas mean free path. For example, a Knudsen number for the flow field could be defined by replacing the particle diameter in Eq. (2.1) with a characteristic reactor length scale. For small particles, it is frequently the case that the flow regime can be considered continuum while the fluid-particle interaction is characterized as free-molecular.

b C is removed from Eq. (2.4) and the expression $1/C$ replaces "1" as the first term on the right-hand side of Eq. (2.5).

c In practice, non-Stokesian effects can be neglected when the particle Reynolds number is less than about 0.3.

d For polyatomic gases, Talbot et al. [17] recommend the use of the 'translational' thermal conductivity, which is given by simple kinetic theory as $k_g = (15/4)\mu R/M$.

e The gas conductivity does play a role, but has been eliminated from Eq. (2.15) by the expression in footnote d.

f The particle stopping distance, τU_0, is defined as the distance a particle would travel before stopping if injected into a quiescent fluid at an initial velocity of U_0.

g The requirement that the characteristic time for particle diffusion is much longer than the particle characteristic time, $t >> \tau$, is also essential to the development of the basic equations of particle transport by Brownian motion (see Ref. [13], Section 35).

h The dilute mixture approximation is certainly valid for simulations of commercial semiconductor process tools, as the particle concentrations are typically controlled to very low levels.

i In the event that the particle axial velocity becomes zero, or begins to move away from the wafer, then the trajectory calculation is terminated and the efficiency is set to zero.

j When inertia is neglected, the particles enter the reactor with the mean gas velocity so that $V_{po} = -U_0$. Also, note that U_0 is the magnitude of the face velocity, and so is positive.

k The inclusion of particle interception effects is somewhat overkill, as we have neglected wafer surface roughness/structure which is likely characterized by dimensions similar to particle sizes.

l However, a study by Collins et al. [37] suggests that some particles might retain a few residual charges (positive or negative) in the afterglow.

m Deposition by interception, due to the finite size of the particle, is neglected. To account for interception requires that the boundary conditions be given as $n(z = r_p) = n(S - r_p) = 0$ where r_p is the particle radius.

n Note that $Re = 8$ is beyond the range over which the analytical approximation was found to be accurate, but the analytical result is used here beyond its range to qualitatively investigate any Reynolds number dependencies.

o An alternate assumption is that the particles follow streamlines, in which case the particle concentration is constant across the tube inlet so that the local flux of particles through the tube depends on radial position.

p A further implication of Eq. (2.70) is that for large values of V_{po}/U_0 a more appropriate choice for the characteristic velocity in defining particle Stokes number would have been V_{po}.

q Note that although $Re = 8$ in Figure 2.10b and $Re = 0$ in Figure 2.22, Reynolds number effects are small.

r The critical Stokes number becomes less dependent on the velocity ratio for small U_{jet}/U_0.

s It is impossible to eliminate inertial effects, as one can always imagine a particle large enough so that it will impact.

t For higher pressures and/or larger particle diameters, the quadratic relationship between pressure and critical diameter fails as the free-molecule expression for Stokes number given by Eq. (2.23) becomes inaccurate. In the continuum limit, the particle relaxation time becomes independent of pressure and proportional to diameter squared, so that a fourfold pressure increase would result in a twofold increase in $d_{p,crit}$. Thus, the influence of pressure on $d_{p,crit}$ is most pronounced in the free-molecule regime.

u Since the thermophoretic drift velocity is independent of particle diameter (Eq. (2.31)) while the settling velocity is proportional to diameter, for small particles thermophoretic deposition dominates and the efficiency becomes constant.

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